

Results — revised submission — enhanced style

1 Datasets

#	Name	N	P	K	MeanIR	CVIR
1	GpositivePseAAC	519	440	4	3.86	1.15
2	Emotions	593	72	6	1.48	0.16
3	Scene	2407	294	6	1.25	0.11
4	PlantPseAAC	978	440	12	6.69	0.68
5	HumanPseAAC	3106	440	14	15.29	1.05
6	Yeast	2417	103	14	7.20	1.82
7	birds	645	260	19	5.41	0.80
8	medical	978	1449	45	222.37	0.72
9	enron	1702	1001	53	10.45	0.80

Table 1: Datasets used in the revised experiments. N , P , K are the number of instances, features, labels. MeanIR and CVIR report per-label imbalance (higher = more imbalanced).

Pending follow-ups (to fill before final submission)

- **ECC baseline on birds (RF + LGBM).** The `birds` folders contain only PA/PR/BR/CC/CLR — ECC has not been run. Method slot 12 currently shows as a gap on every `birds` panel. Re-run with `scripts/train_extra_baselines.py --dataset birds --algorithm ecc` for both `--base_learner rf` and `--base_learner lgbm`.
- **LGBM run on enron.** Sequentially training (2–3h per task due to $K=53$ pairwise classifiers); will be merged into `full_enron_split_lgbm_summary/` when complete and the LGBM aggregate panels regenerated.
- **Extended PartialAbstention metrics.** The evaluator has been extended with 11 new PA metrics (`jaccard_pa`, `{example,macro,micro}-{precision,recall,f1}_pa`, `mfrd_pa`, `afrd_pa`). Existing runs were summarised with the previous metric set; the new columns will populate automatically on the next end-to-end training+evaluation pass.

2 Method encoding

For every panel that follows, the x-axis encodes the classifier as one of 12 methods, grouped into three blocks:

Block A — Order-based predictors (ours, indices 1–8). Each name has the form `<order>-<target>-<height>`; $order \in \{\text{PA} = \text{partial order}, \text{PR} = \text{pre-order}\}$; $target \in \{\text{H} = \text{Hamming}, \text{S} = \text{Subset 0/1}\}$; $height \in \{2, \infty \text{ (omitted in the label)}\}$.

- 1 = **PA-H-2**: partial order, Hamming target, height ≤ 2 .
- 2 = **PA-H**: partial order, Hamming target, unrestricted height.
- 3 = **PA-S-2**: partial order, Subset 0/1 target, height ≤ 2 .
- 4 = **PA-S**: partial order, Subset 0/1, unrestricted.
- 5 = **PR-H-2**: pre-order, Hamming, height ≤ 2 .
- 6 = **PR-H**: pre-order, Hamming, unrestricted.
- 7 = **PR-S-2**: pre-order, Subset 0/1, height ≤ 2 .
- 8 = **PR-S**: pre-order, Subset 0/1, unrestricted.

Block B — Standard MLC baselines from the original paper (indices 9–11).

- 9 = **BR** (Binary Relevance): K independent binary classifiers, one per label.
- 10 = **CC** (Classifier Chain): sequential chain that uses previous labels as features.
- 11 = **CLR** (Calibrated Label Ranking): pairwise label ranking with a calibration label that separates relevant from irrelevant.

Block C — Extended baseline added for the revision (index 12).

- 12 = **ECC** (Ensemble of Classifier Chains): averages over multiple CCs with random chain orders.

Reading the panels. Each panel plots one dotted line with markers per noise level. **Noise is encoded by warm-hue intensity** (ColorBrewer YlOrRd): yellow = clean ($\alpha = 0.0$), dark red = noisy ($\alpha = 0.3$): ■ $\alpha = 0.0$, ■ $\alpha = 0.1$, ■ $\alpha = 0.2$, ■ $\alpha = 0.3$. Below the numeric x-axis (1–12), a second tier prints the short method name so the reader can identify each classifier without referring back to this section. *PartialAbstention* and *ScoreVector* panels show only indices 1–8. A vertical dashed line at $x = 8.5$ separates the order-based block (1–8) from the standard MLC baselines (9–12). The y-axis is zoomed to the actual data range to keep close-clustered curves visually separable. Missing methods show as gaps in the line (with a small grey “x” on the axis when the method has no data at all for that panel).

3 Main results: RF base learner, aggregated

RF • Aggregate • BinaryVector — Standard MLC predictions

RF • Aggregate • PartialAbstention — Partial abstention

RF • Aggregate • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

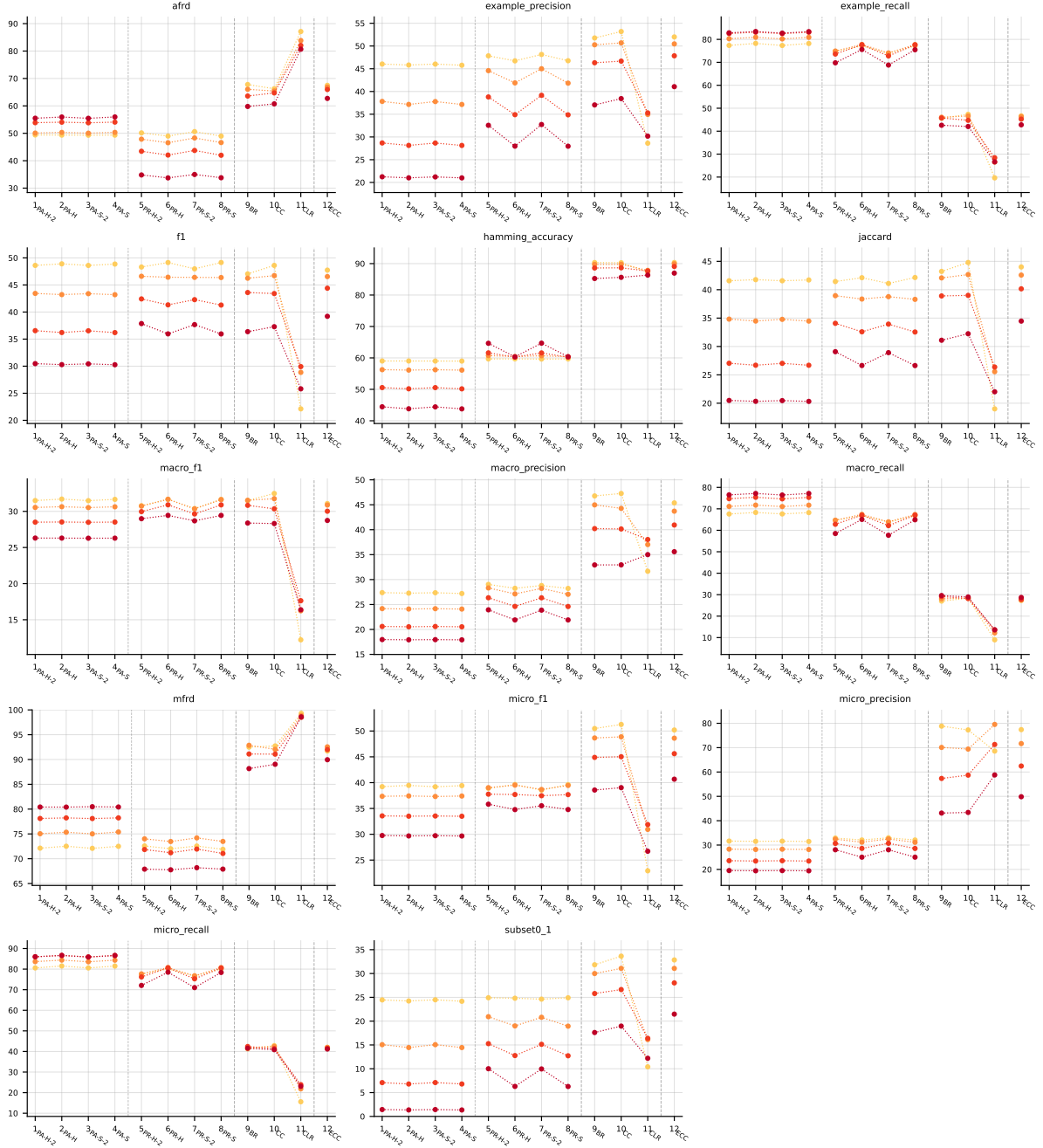


Table 2: Standard MLC predictions (BinaryVector), RF base learner, averaged across datasets. Method indices and line colors follow the encoding paragraph above.

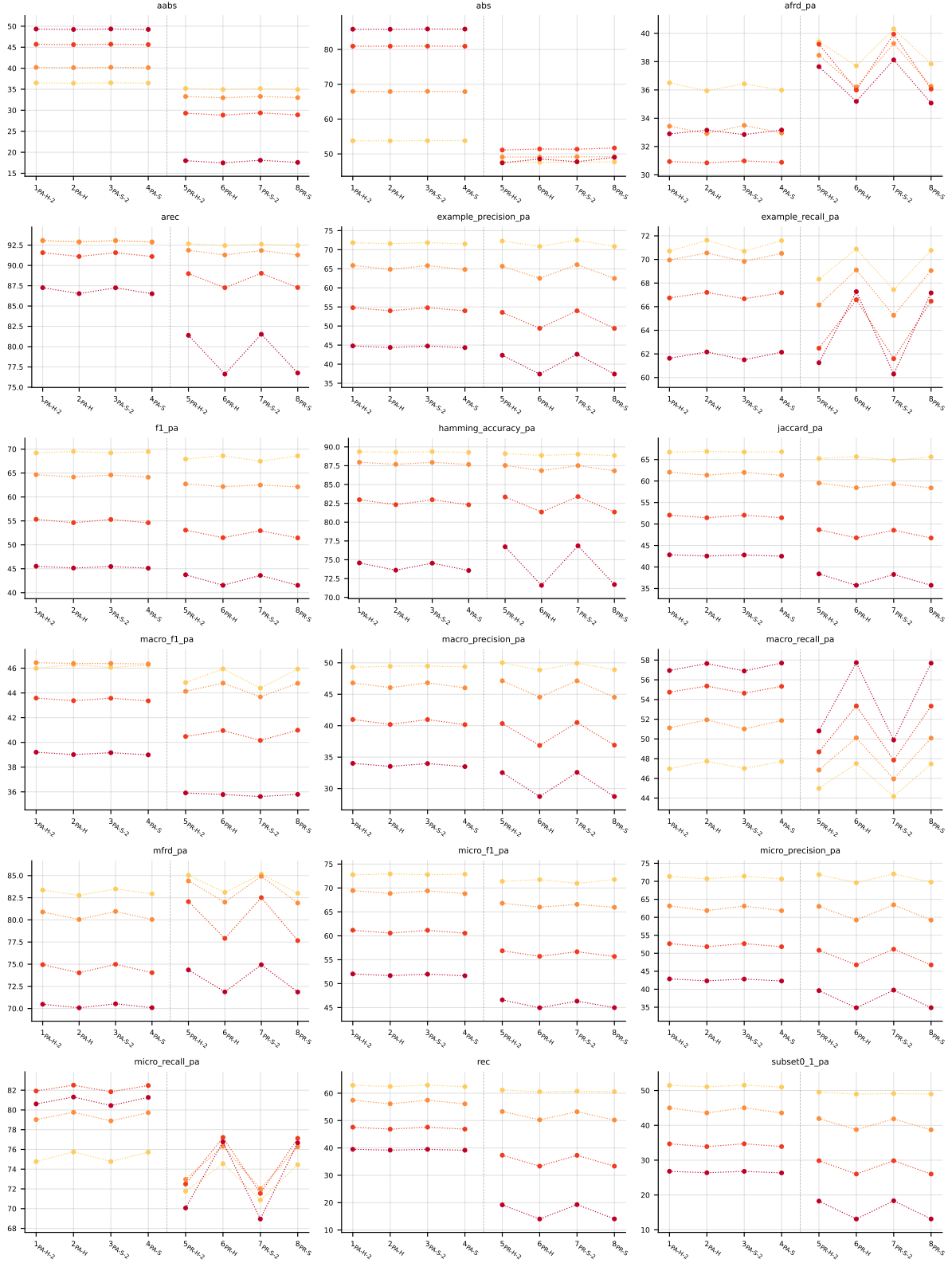


Table 3: Partial abstention (PartialAbstention), RF base learner, averaged across datasets. Method indices and line colors follow the encoding paragraph above.

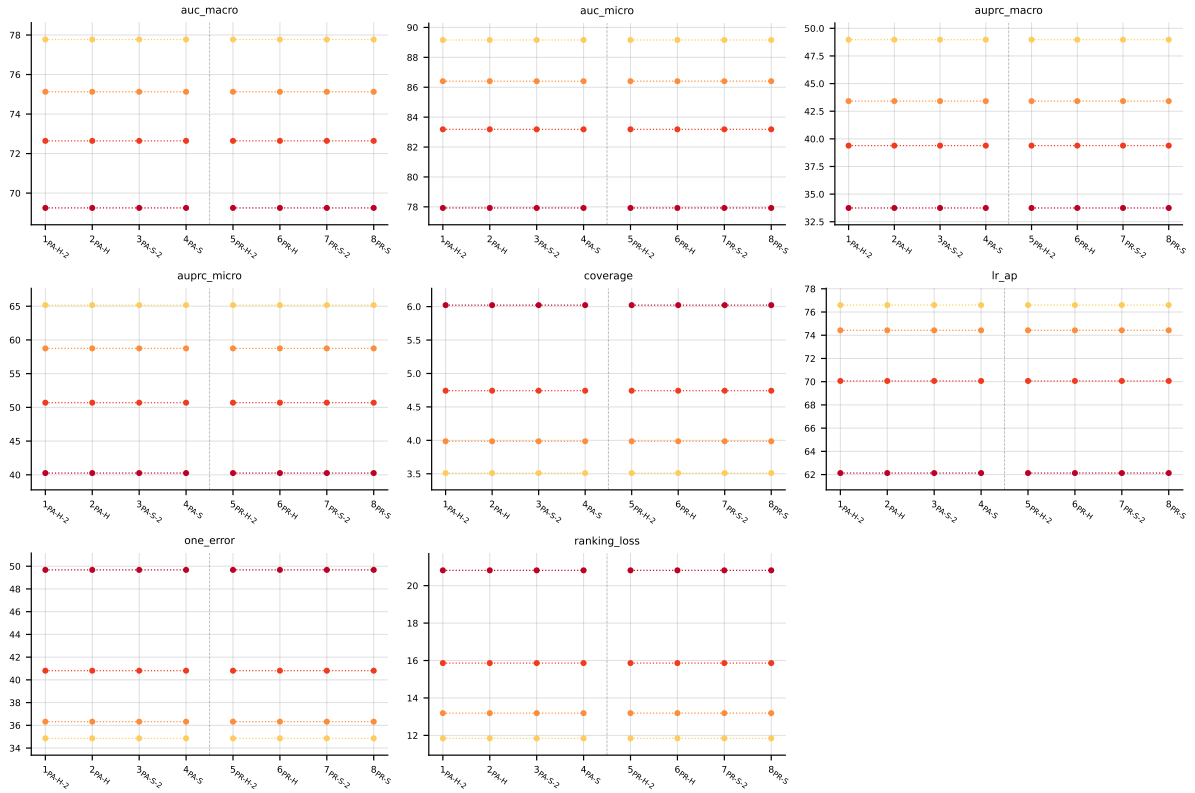


Table 4: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector), RF base learner, averaged across datasets. Method indices and line colors follow the encoding paragraph above.

4 Abstention vs standard MLC

Why this section exists. The paper’s central claim is that an order-based predictor can *abstain* on labels it is unsure about and, on the labels it does predict, achieve higher quality than a standard MLC method that is forced to predict every label. The two chart families below make that claim visible.

Terminology. ‘*Standard MLC*’ (a.k.a. **BinaryVector** / BV) means each method outputs a 0/1 prediction for every one of the K labels. ‘*With abstention*’ (a.k.a. **PartialAbstention** / PA) means an order-based predictor may return $\perp$ (abstain) on some labels; the PA quality metrics (**f1_pa**, **jaccard_pa**, ...) are computed only over the non-abstained labels. Only the 8 PA/PR methods (indices 1–8) can abstain; BR/CC/CLR/ECC cannot.

Reading the coverage-risk chart. Each point is one (PA/PR method, noise level α) pair. The x-axis is the abstention rate (fraction of labels the method skipped); the y-axis is the quality metric on the labels that *were* predicted (higher is better). Marker shape encodes method, colour encodes noise level. The dashed grey line is the best standard-MLC baseline (BR / CC / CLR / ECC) on the BV quality metric — it is the bar that “no abstention” has to clear. **Any point above the dashed line is direct evidence that abstaining improved quality beyond what any standard MLC method achieves.** Points further right traded coverage for that gain.

Reading the paired-bars chart. Same data, different view. The four sub-panels correspond to the four noise levels. Inside each sub-panel: 12 method positions, two bars per position — blue is the standard-MLC quality (**f1**, no abstention), warm-colour is the with-abstention quality on retained labels. Only the 8 PA/PR methods (1–8) have a warm bar; the standard MLC baselines (9–12) only have a blue bar by construction. The green “+ $x.y$ ” label above each warm bar is the gain in score-points over that method’s own BV bar.

4.1 RF, metric f1_pa — aggregate



Figure 1: RF (f1_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right), averaged across all datasets.

4.2 RF, metric f1_pa — per dataset

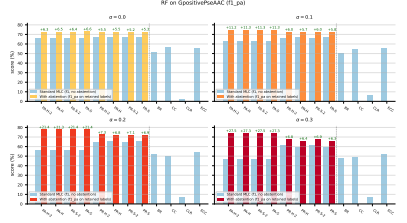
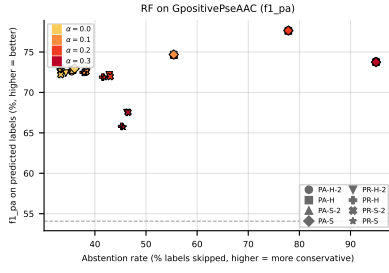


Figure 2: RF on **GpositivePseAAC** (f1_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

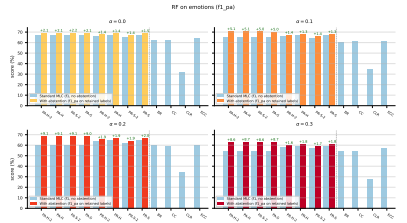
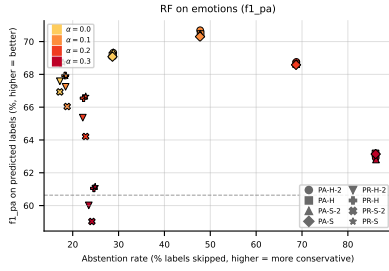


Figure 3: RF on **emotions** (f1_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

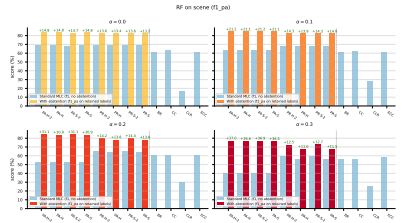
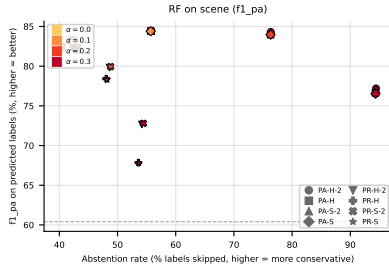


Figure 4: RF on **scene** (f1_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

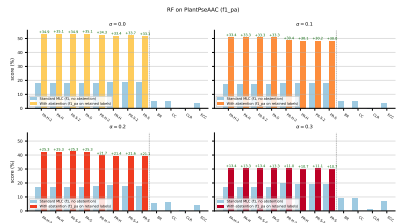
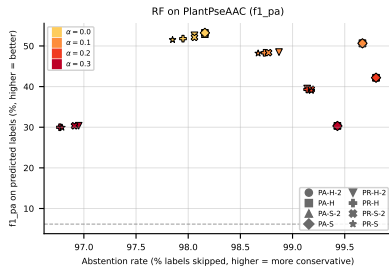


Figure 5: RF on **PlantPseAAC** (f1_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

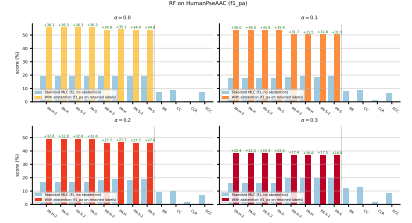
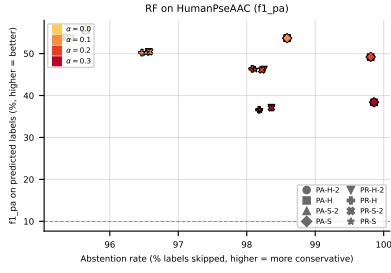


Figure 6: RF on HumanPseAAC (f1_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

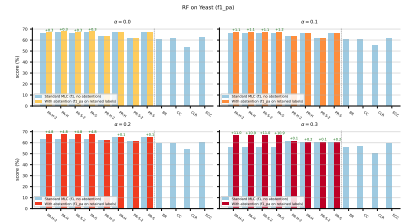
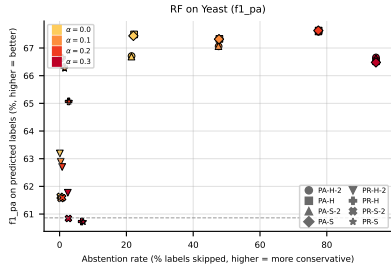


Figure 7: RF on Yeast (f1_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

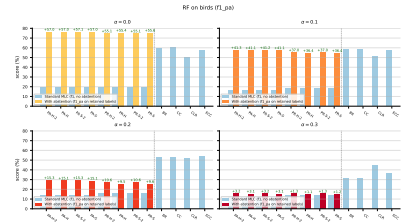
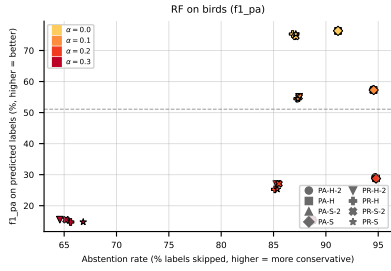


Figure 8: RF on birds (f1_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

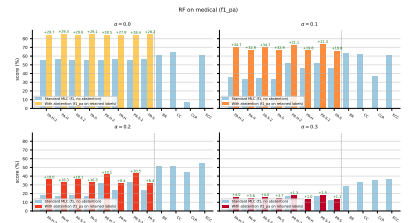
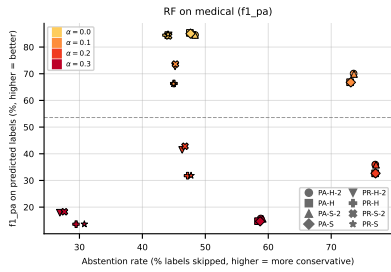


Figure 9: RF on medical (f1_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

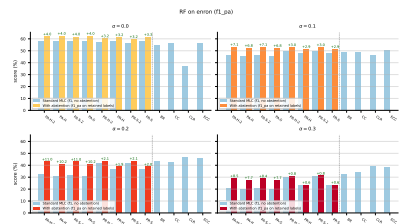
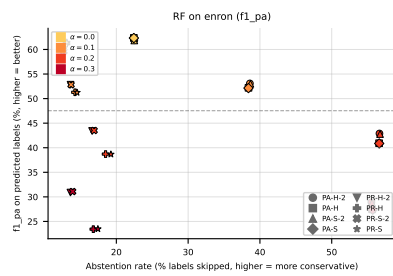


Figure 10: RF on **enron** (f1_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

4.3 RF, metric jaccard_pa — aggregate



Figure 11: RF (jaccard_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right), averaged across all datasets.

4.4 RF, metric jaccard_pa — per dataset



Figure 12: RF on GpositivePseAAC (jaccard_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).



Figure 13: RF on **emotions** (jaccard_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

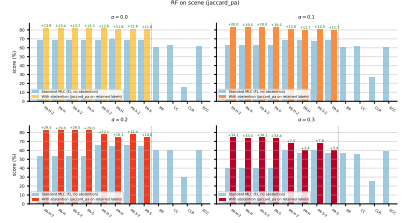
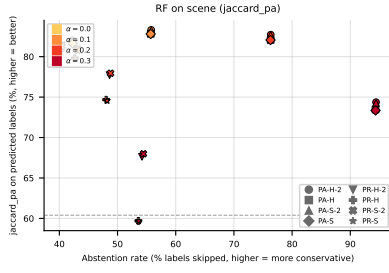


Figure 14: RF on **scene** (jaccard_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

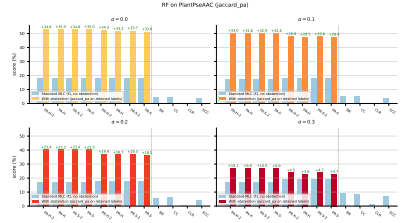
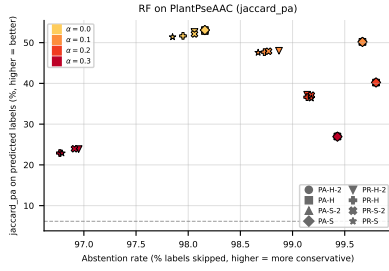


Figure 15: RF on **PlantPseAAC** (jaccard_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

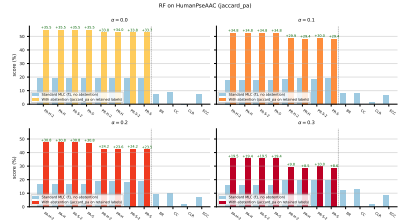
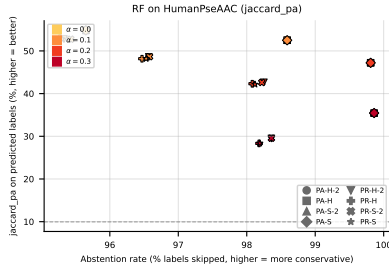


Figure 16: RF on **HumanPseAAC** (jaccard_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

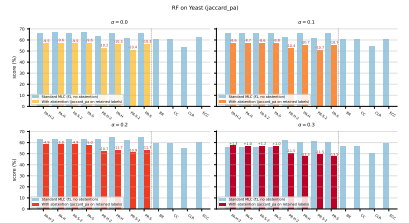
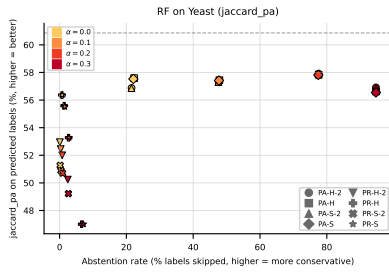


Figure 17: RF on **Yeast** (jaccard_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

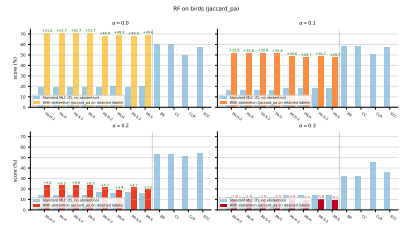
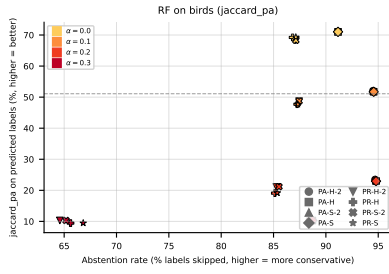


Figure 18: RF on **birds** (jaccard_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

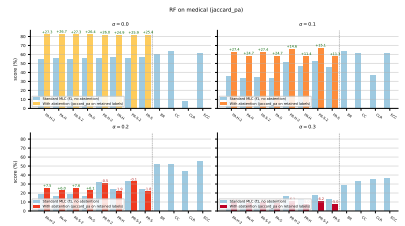
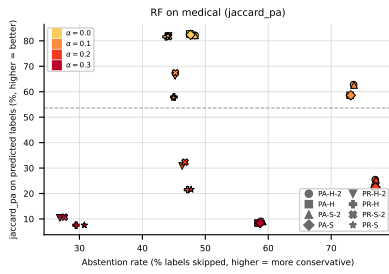


Figure 19: RF on **medical** (jaccard_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

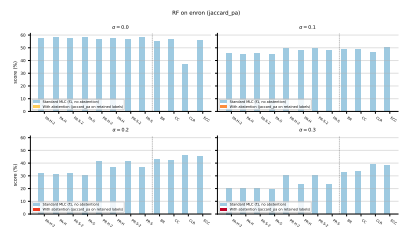
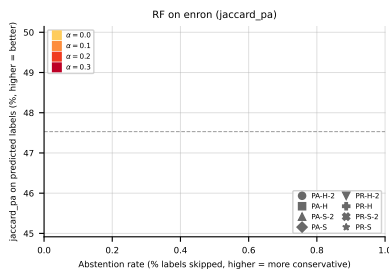


Figure 20: RF on **enron** (jaccard_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

4.5 LGBM, metric f1_pa — aggregate

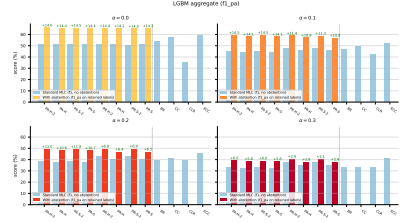
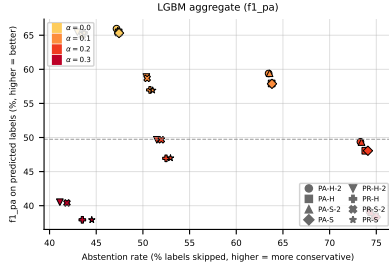


Figure 21: LGBM (f1_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right), averaged across all datasets.

4.6 LGBM, metric f1_pa — per dataset

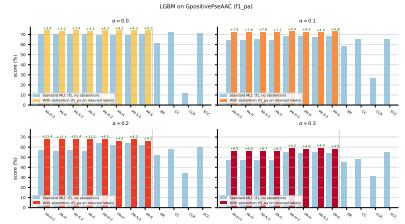
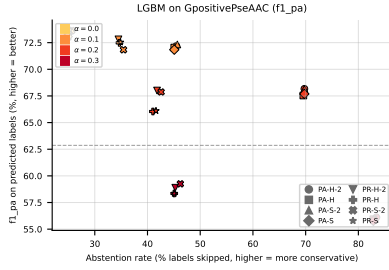


Figure 22: LGBM on GpositivePseAAC (f1_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

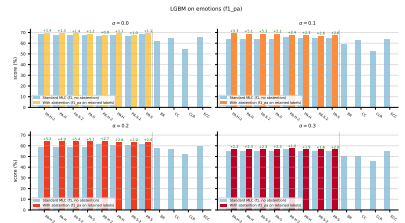
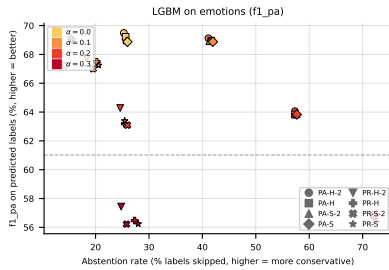


Figure 23: LGBM on emotions (f1_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

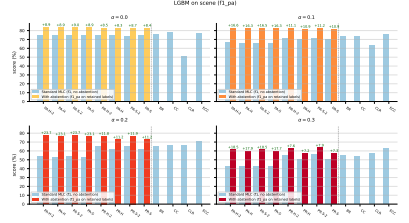
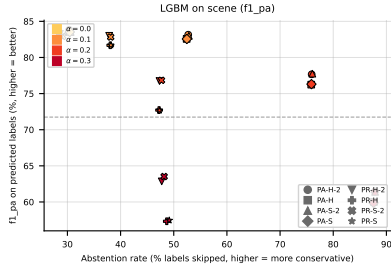


Figure 24: LGBM on **scene** (f1_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

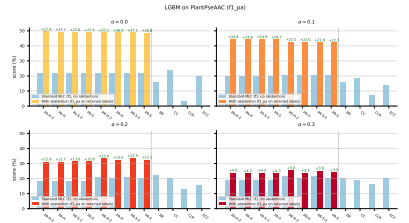
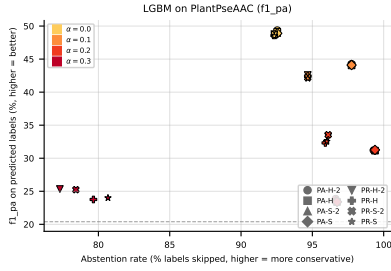


Figure 25: LGBM on **PlantPseAAC** (f1_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

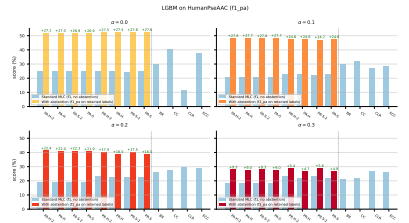
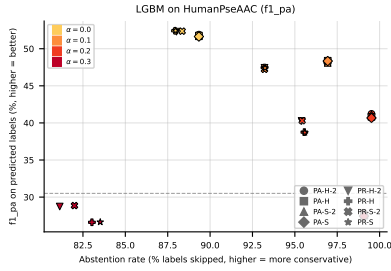


Figure 26: LGBM on **HumanPseAAC** (f1_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

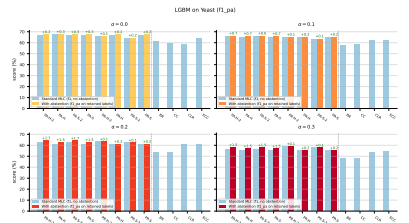
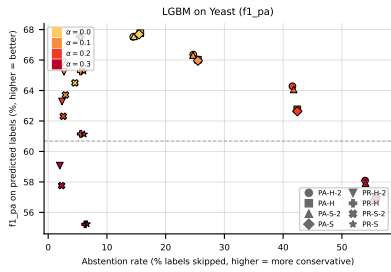


Figure 27: LGBM on **Yeast** (f1_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

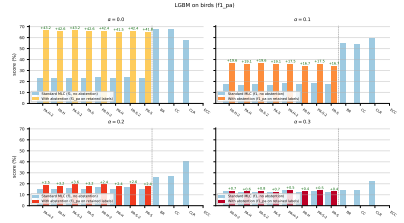
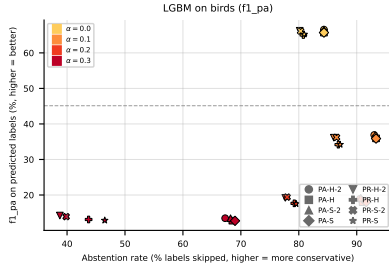


Figure 28: LGBM on birds (f1_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

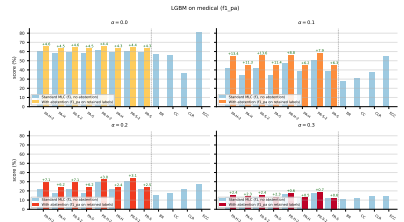
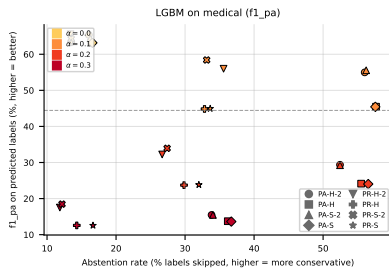


Figure 29: LGBM on medical (f1_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

4.7 LGBM, metric jaccard_pa — aggregate

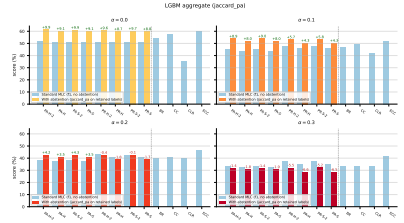
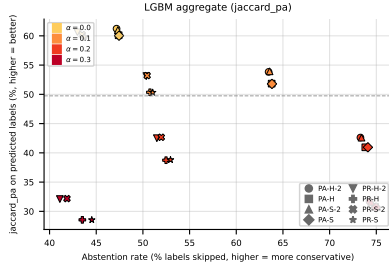


Figure 30: LGBM (jaccard_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right), averaged across all datasets.

4.8 LGBM, metric jaccard_pa — per dataset

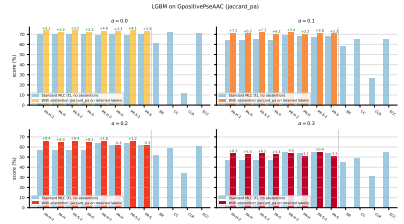
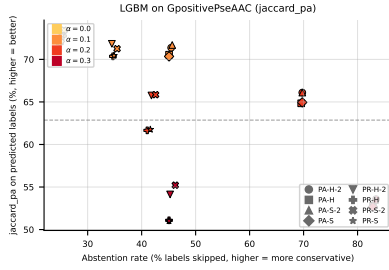


Figure 31: LGBM on GpositivePseAAC (jaccard_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

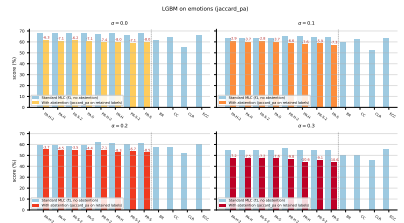
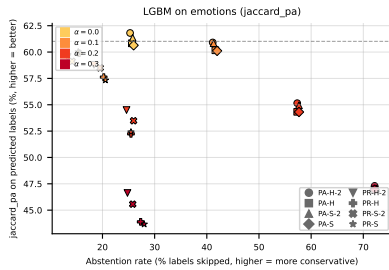


Figure 32: LGBM on emotions (jaccard_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

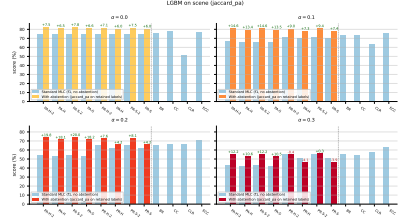
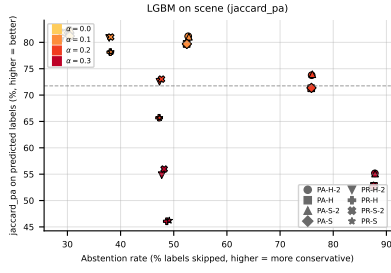


Figure 33: LGBM on **scene** (jaccard_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

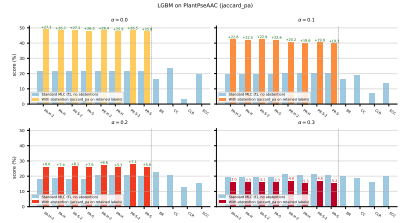
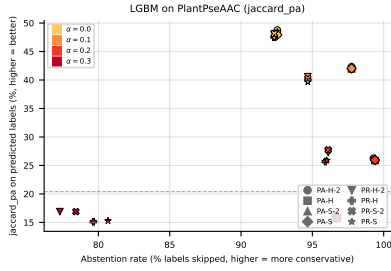


Figure 34: LGBM on **PlantPseAAC** (jaccard_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

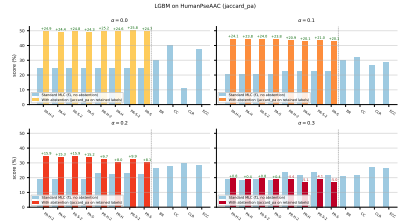
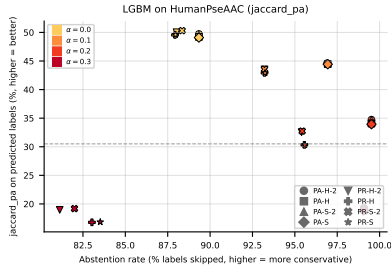


Figure 35: LGBM on **HumanPseAAC** (jaccard_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

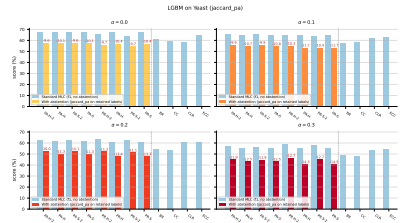
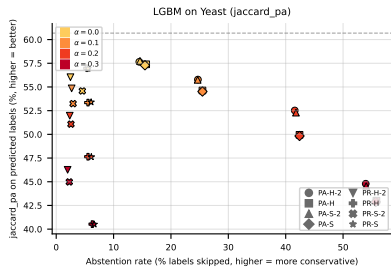


Figure 36: LGBM on **Yeast** (jaccard_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

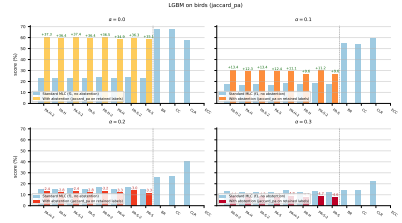
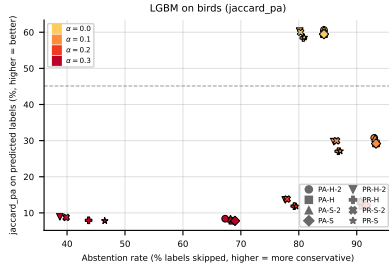


Figure 37: LGBM on **birds** (jaccard_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

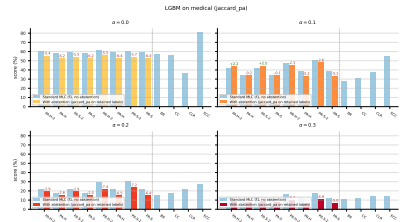
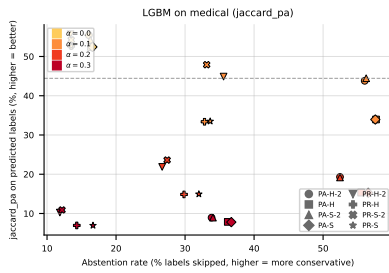


Figure 38: LGBM on **medical** (jaccard_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

A Per-dataset results (RF base learner)

A.1 GpositivePseAAC ($K = 4$)

RF • GpositivePseAAC • BinaryVector — Standard MLC predictions

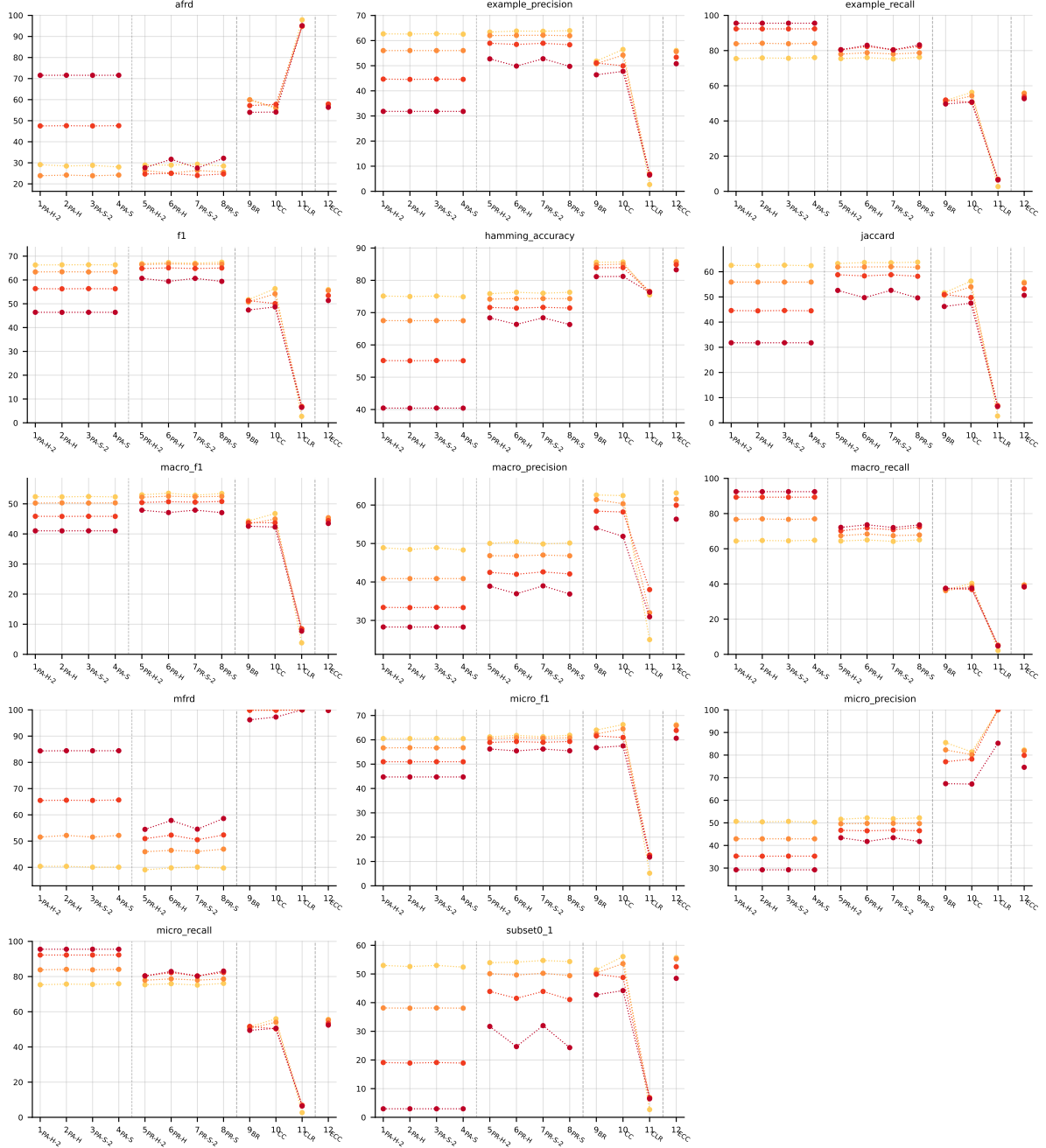


Table 5: Standard MLC predictions (BinaryVector) on GpositivePseAAC (RF base learner).

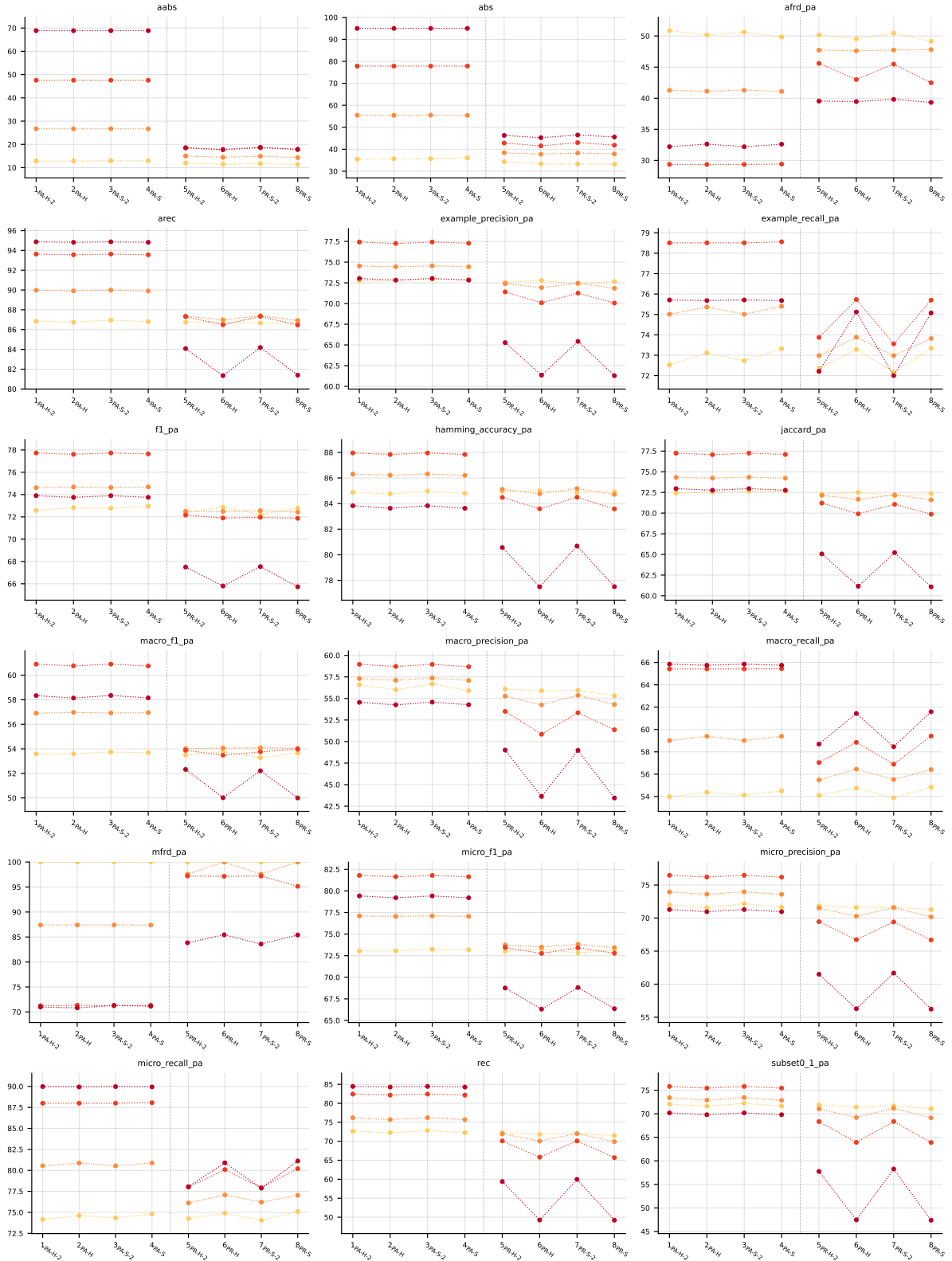


Table 6: Partial abstention (PartialAbstention) on GpositivePseAAC (RF base learner).

RF • GpositivePseAAC • PartialAbstention — Partial abstention

RF • GpositivePseAAC • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

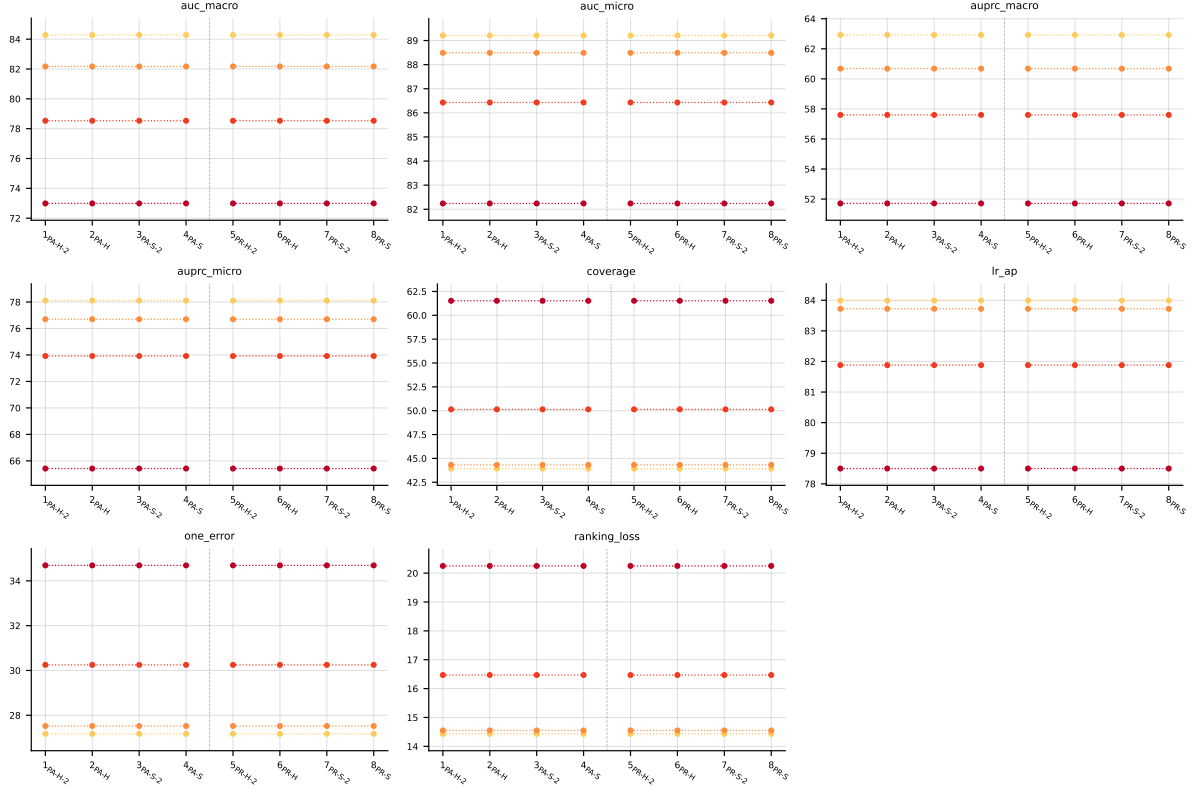


Table 7: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on GpositivePseAAC (RF base learner).

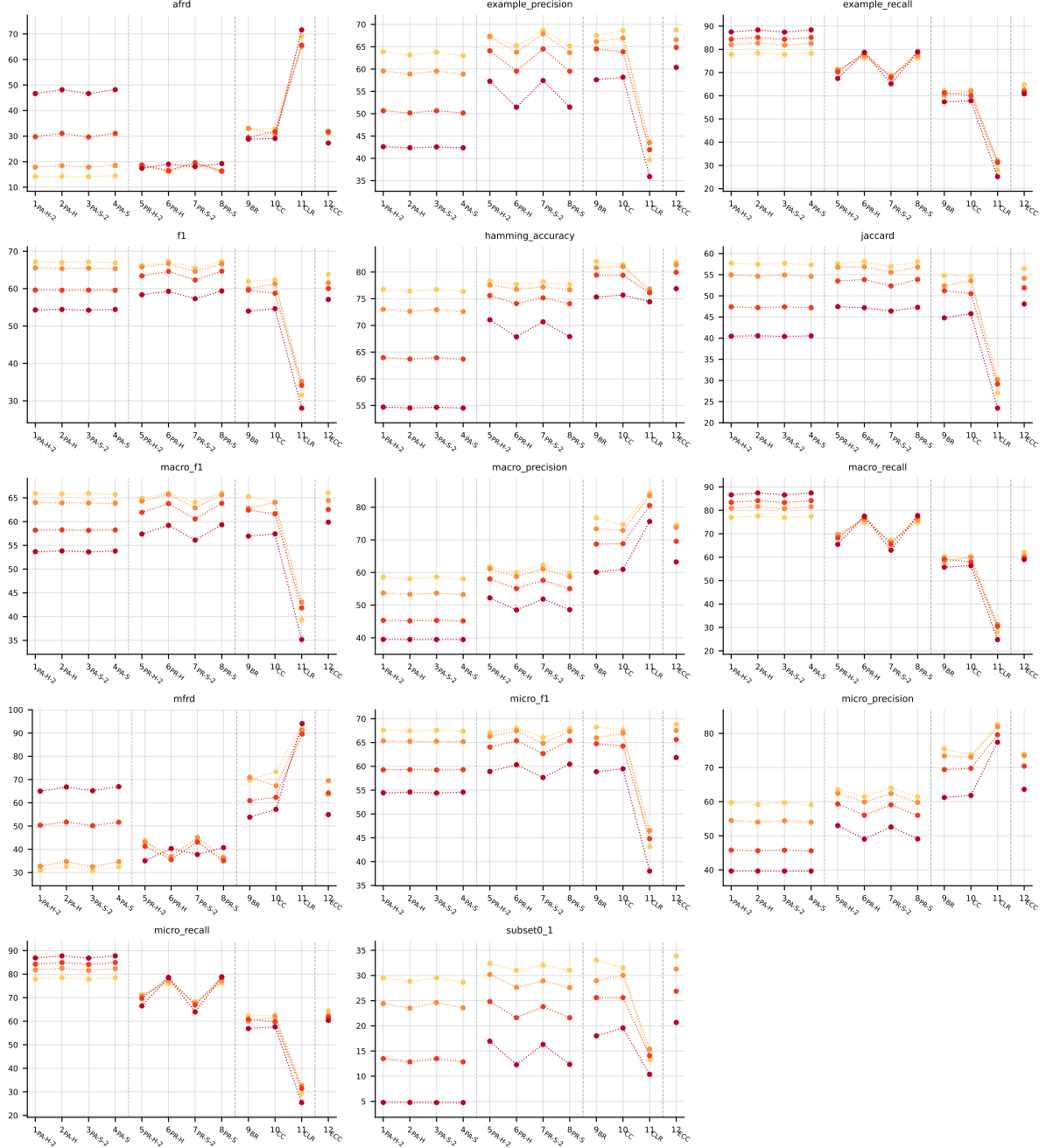


Table 8: Standard MLC predictions (BinaryVector) on emotions (RF base learner).

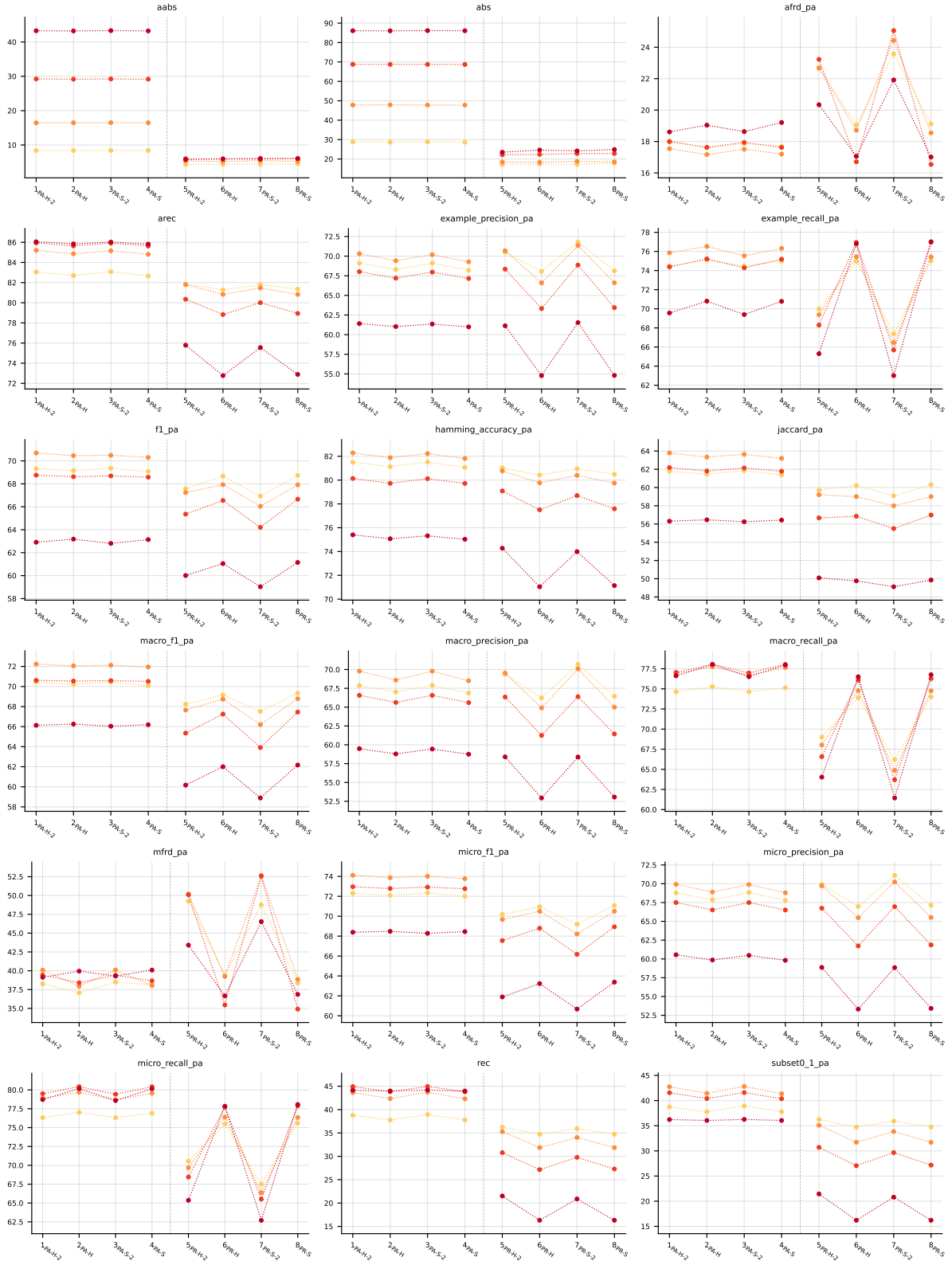


Table 9: Partial abstention (PartialAbstention) on emotions (RF base learner).

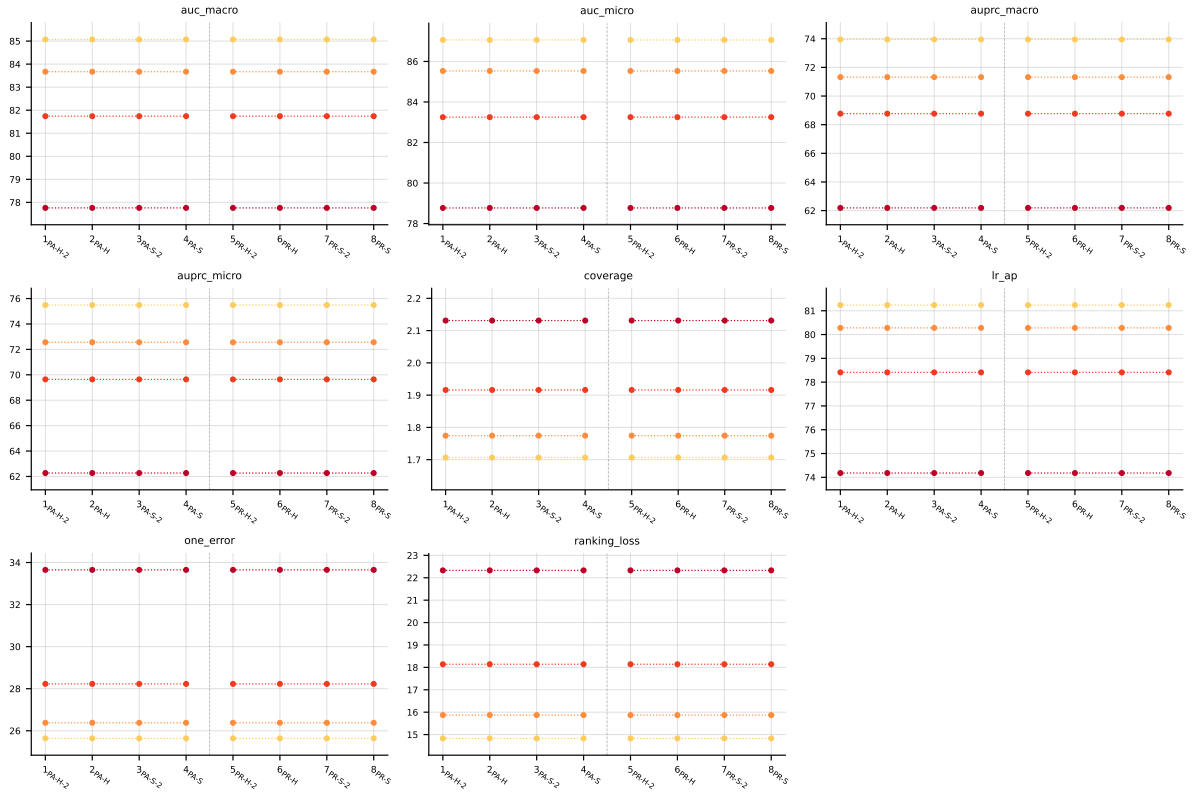


Table 10: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on **emotions** (RF base learner).

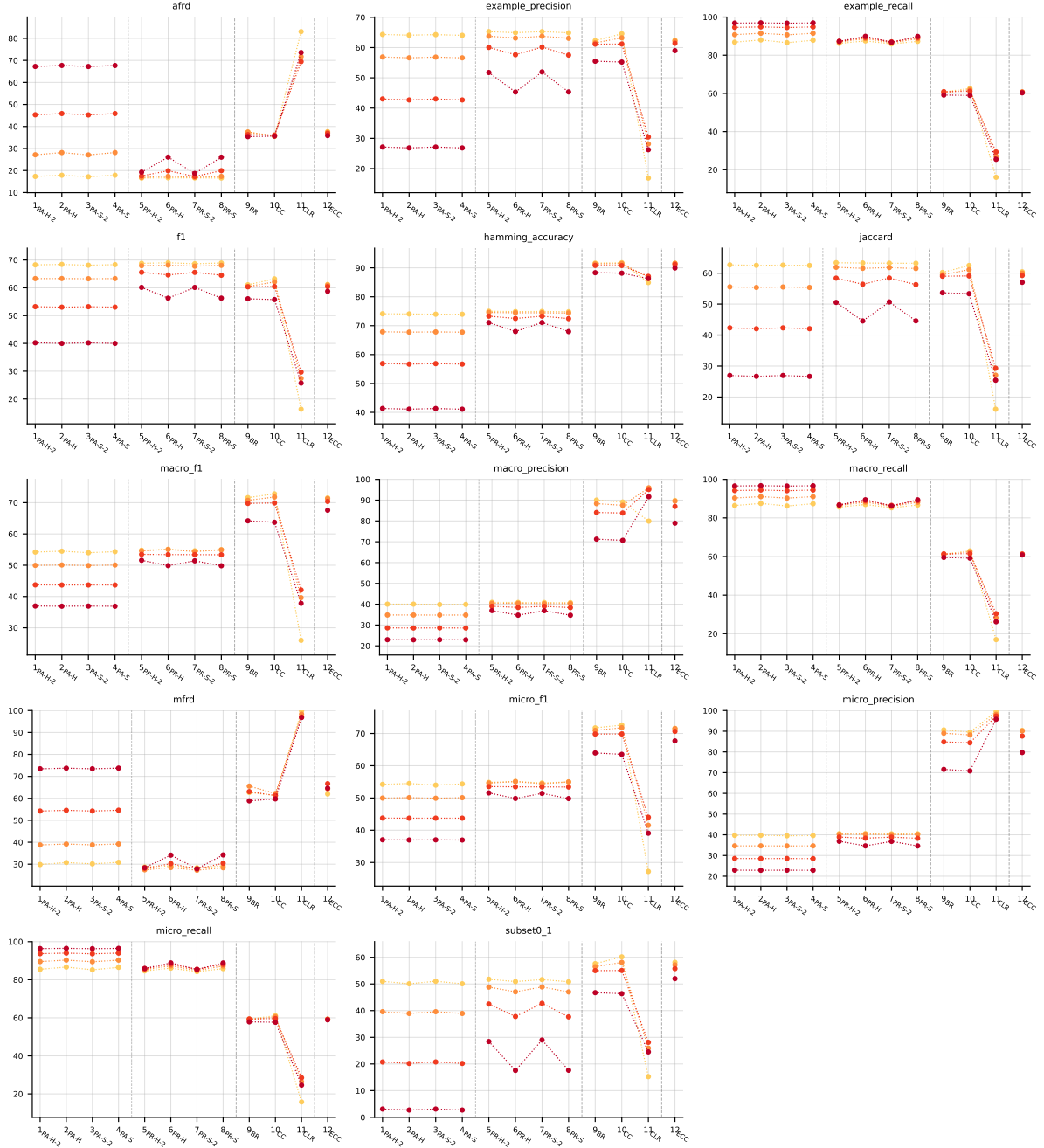


Table 11: Standard MLC predictions (BinaryVector) on **scene** (RF base learner).

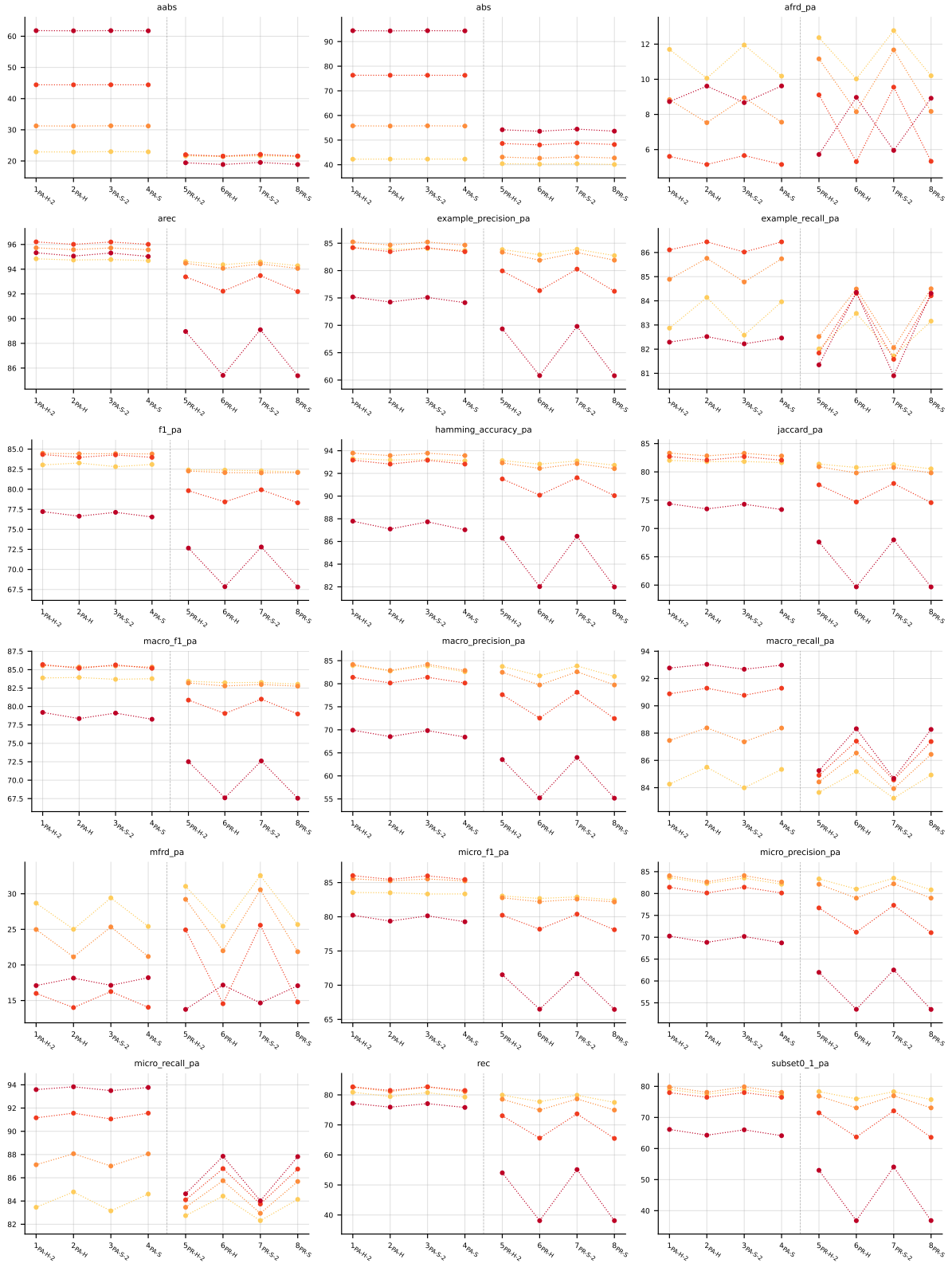


Table 12: Partial abstention (PartialAbstention) on scene (RF base learner).

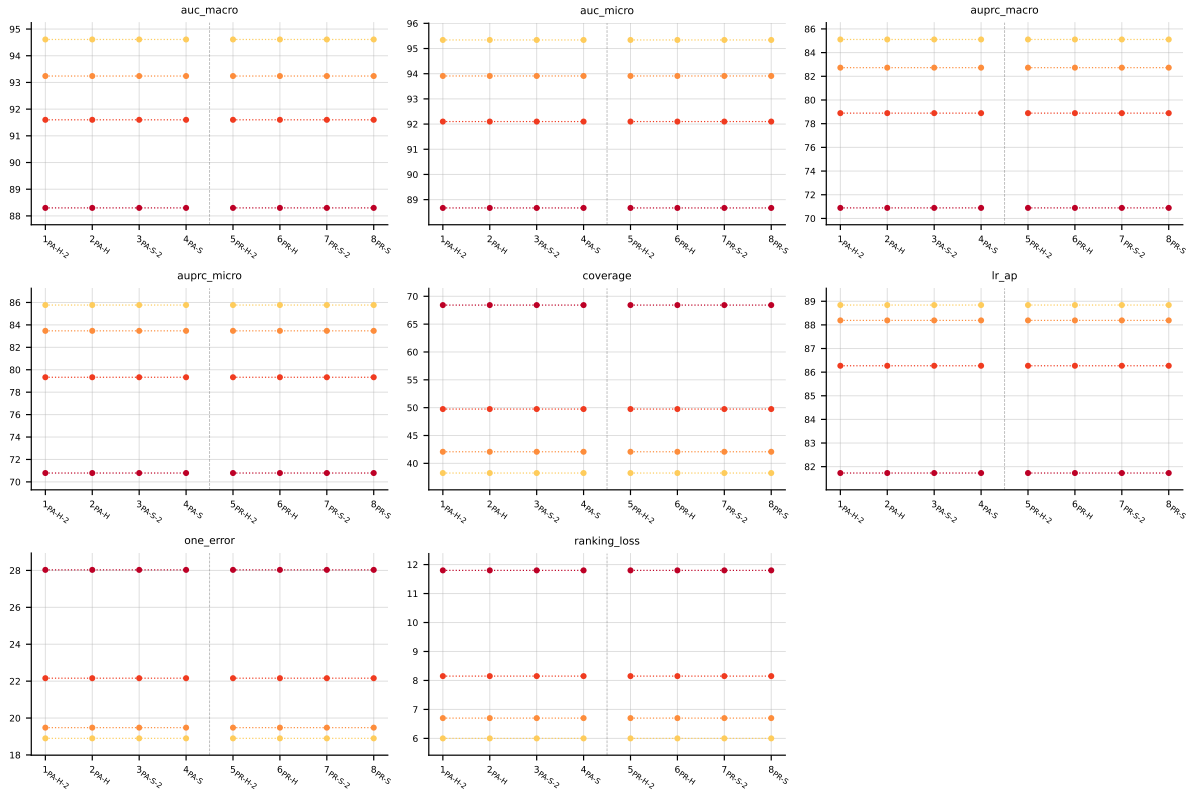


Table 13: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on **scene** (RF base learner).

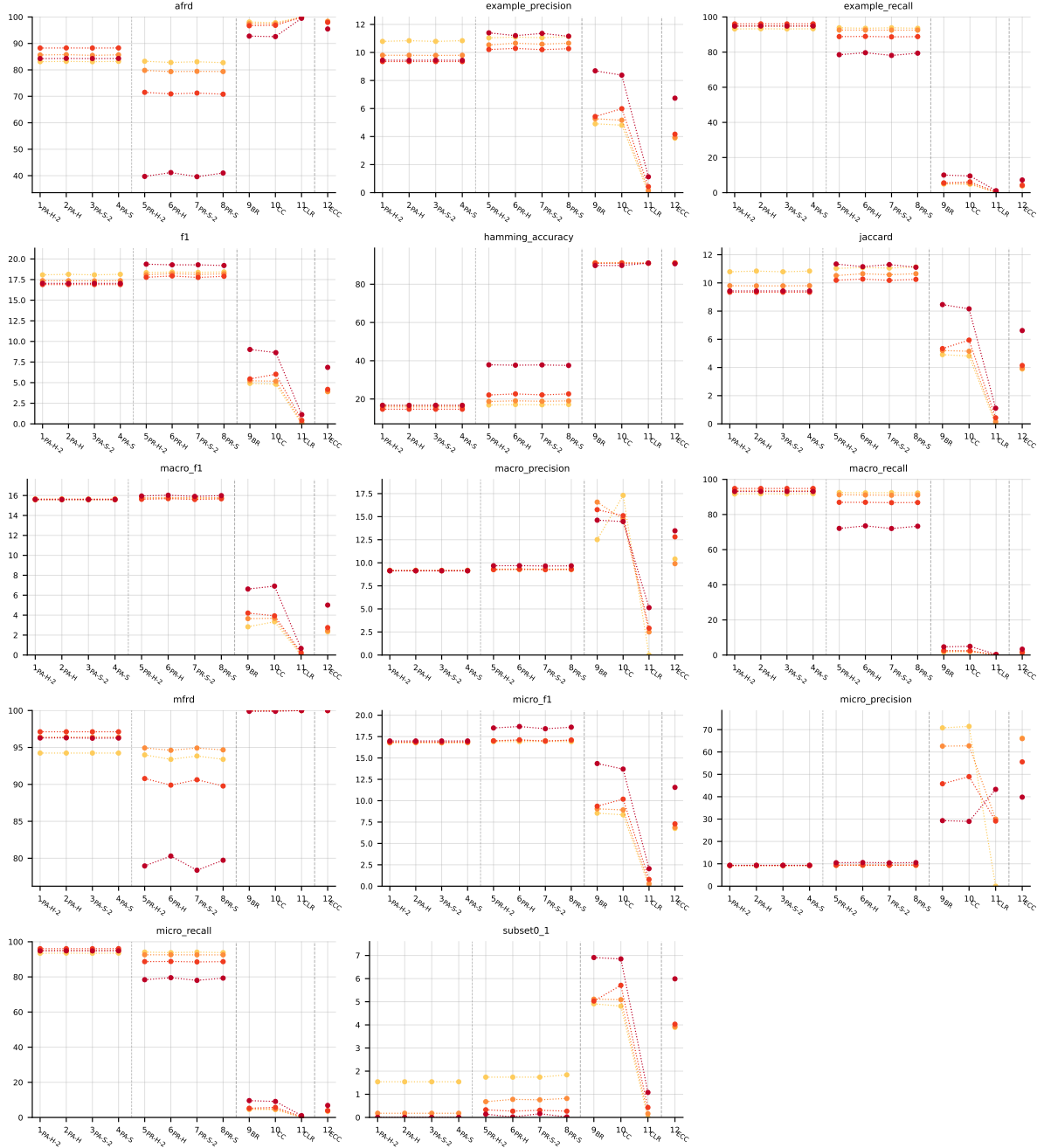


Table 14: Standard MLC predictions (BinaryVector) on PlantPseAAC (RF base learner).

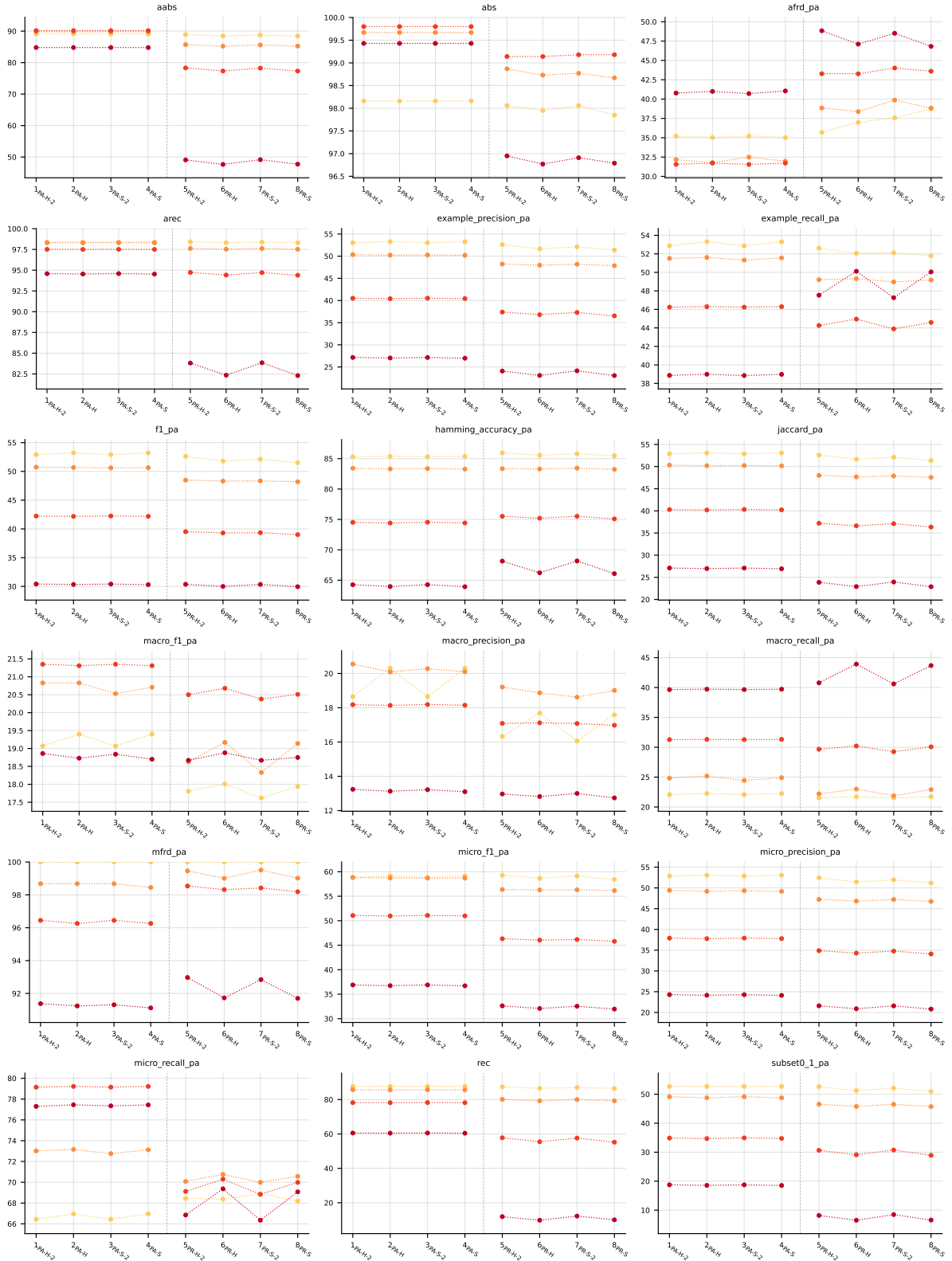


Table 15: Partial abstention (PartialAbstention) on PlantPseAAC (RF base learner).

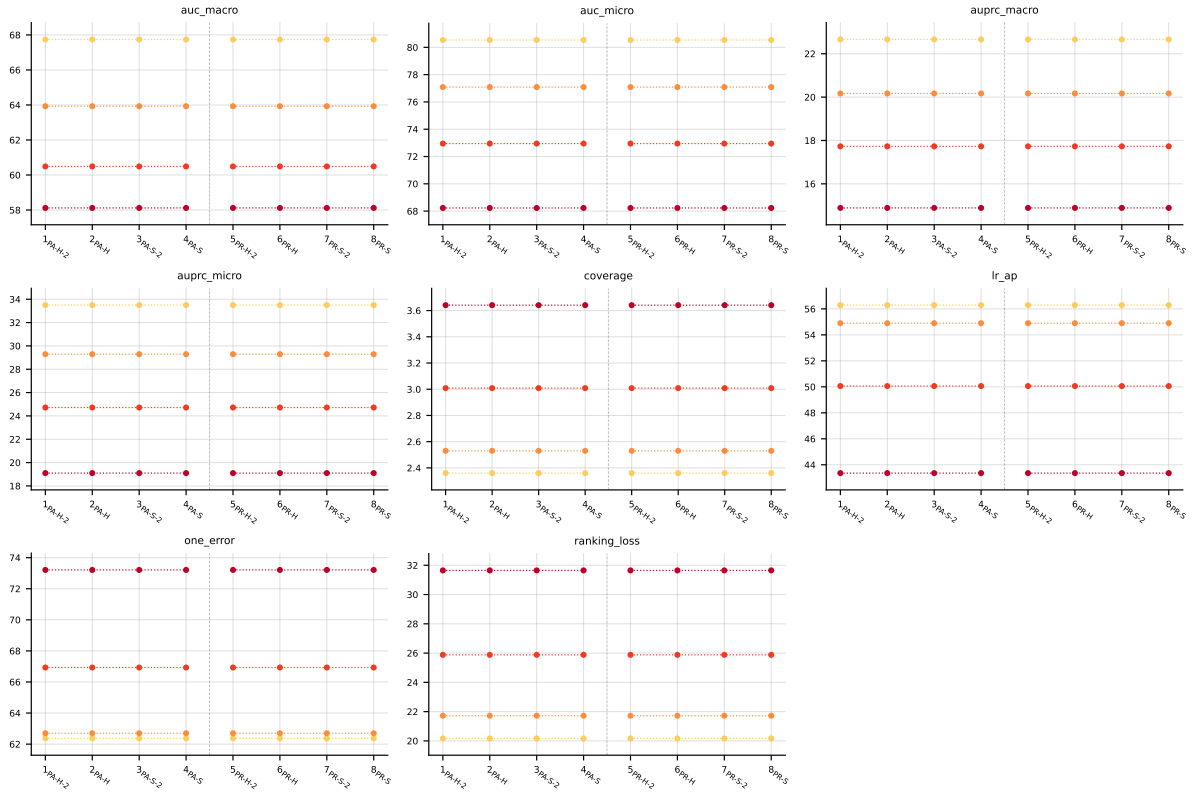


Table 16: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on PlantPseAAC (RF base learner).

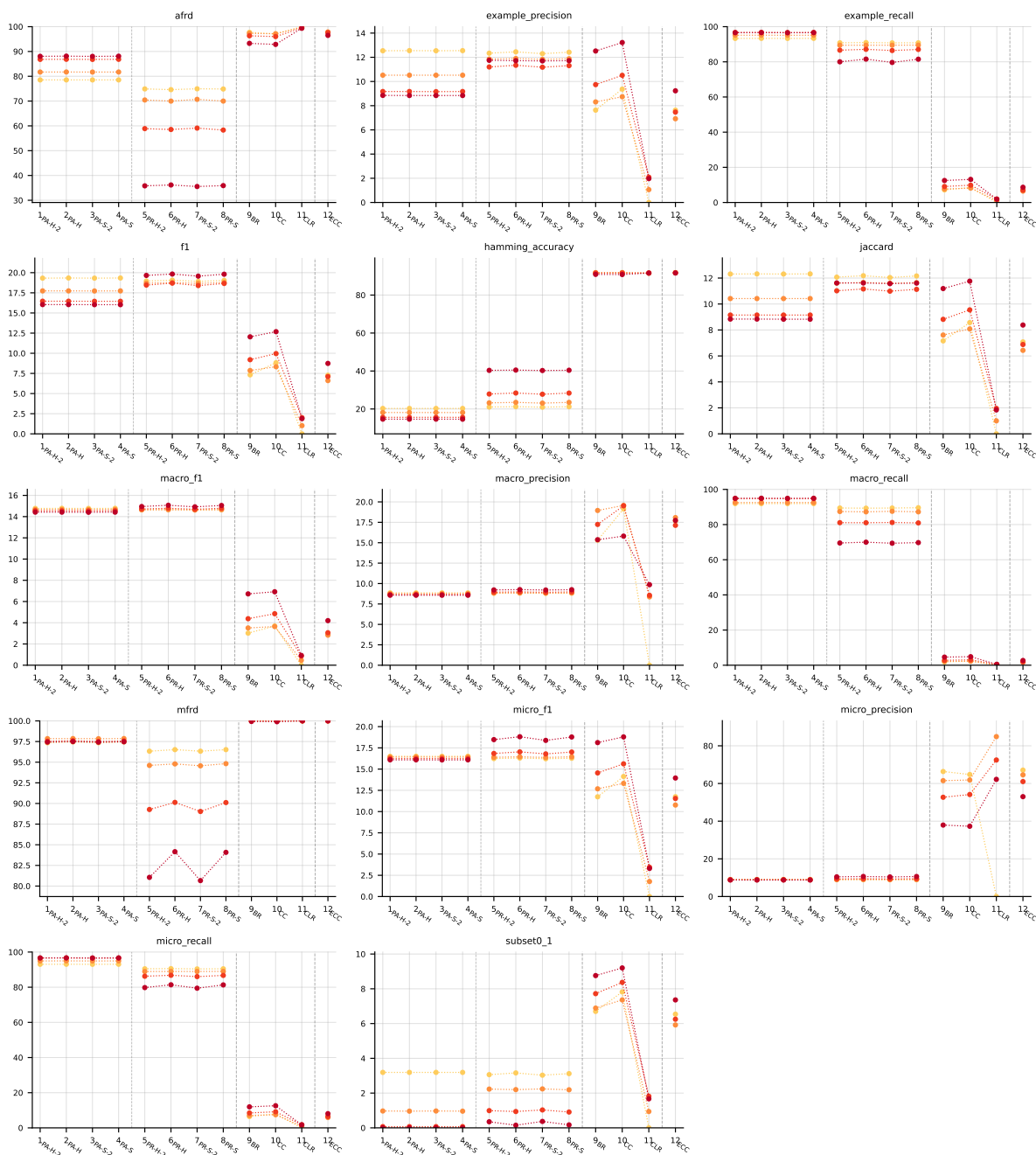


Table 17: Standard MLC predictions (BinaryVector) on HumanPseAAC (RF base learner).

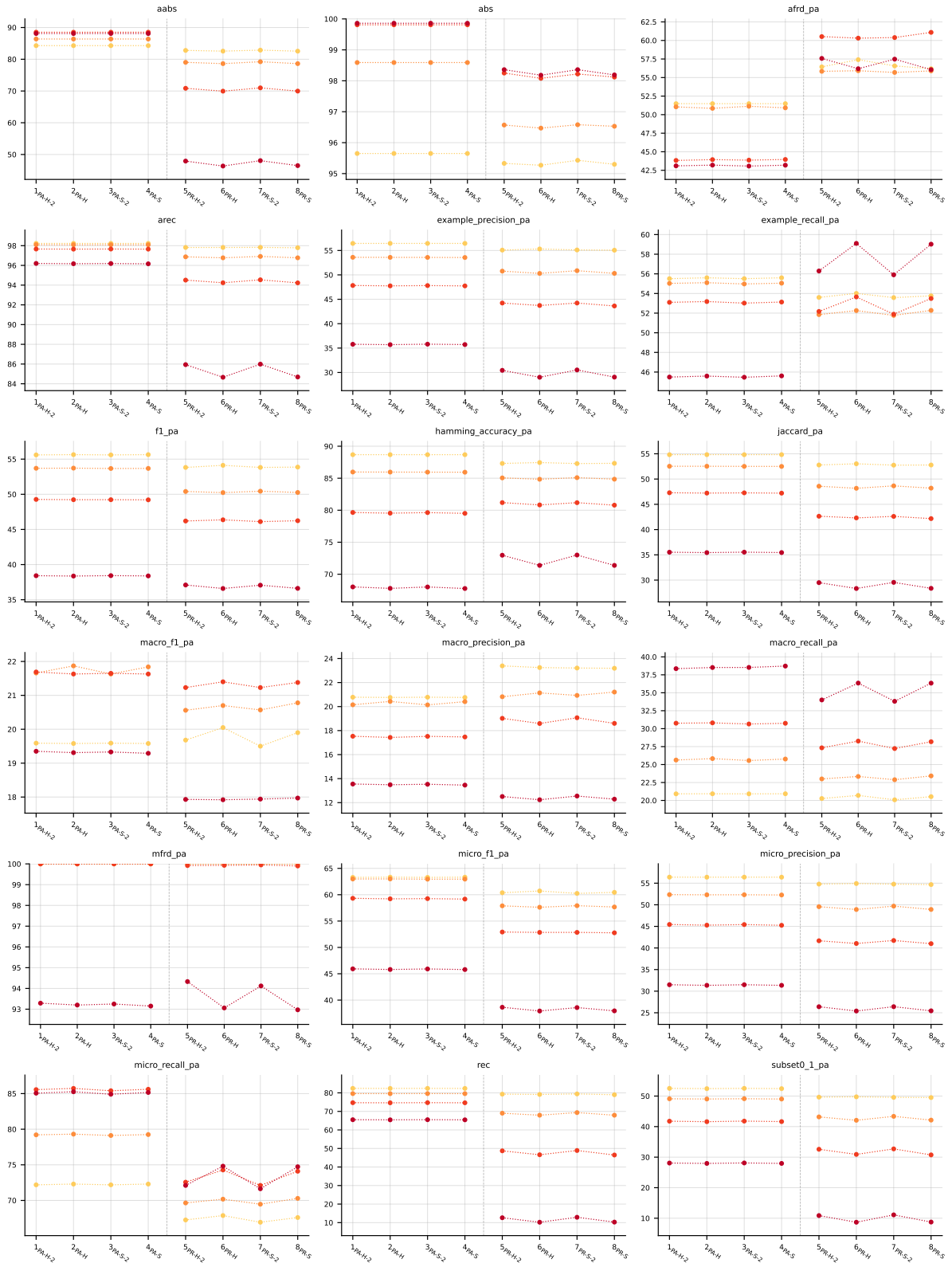


Table 18: Partial abstention (PartialAbstention) on HumanPseAAC (RF base learner).

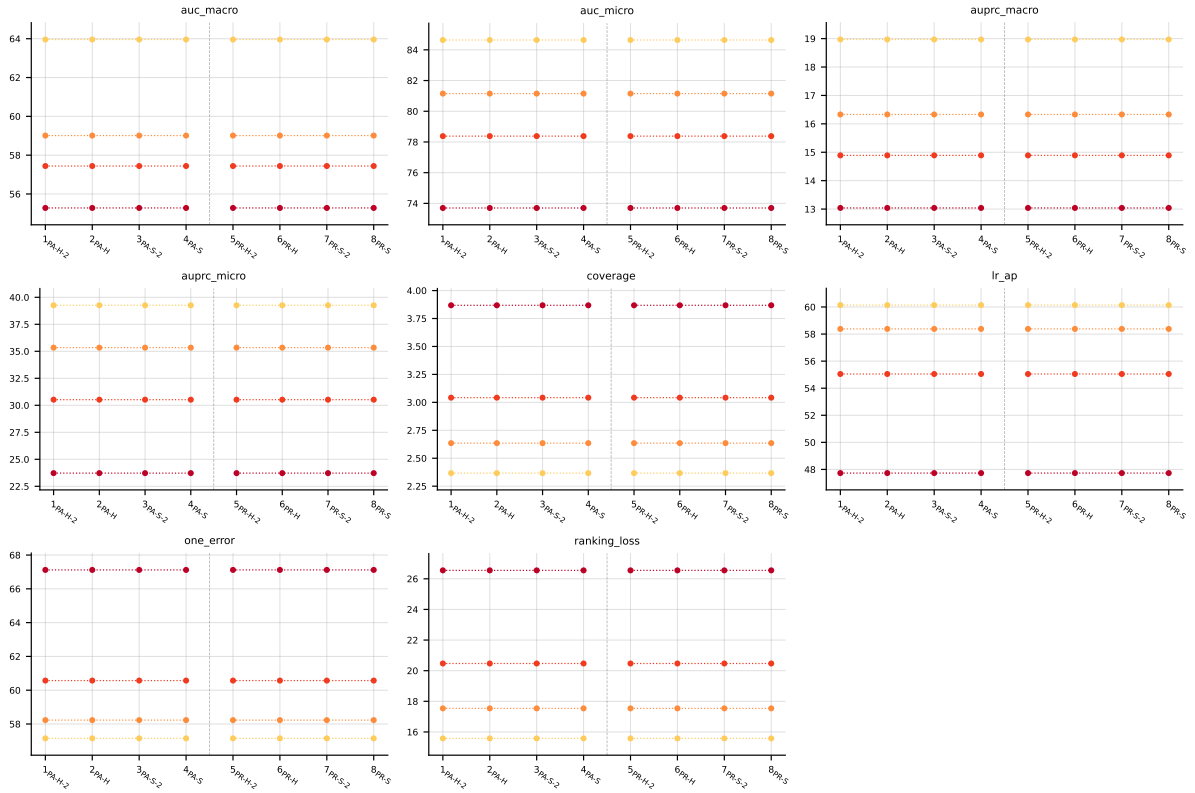


Table 19: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on HumanPseAAC (RF base learner).

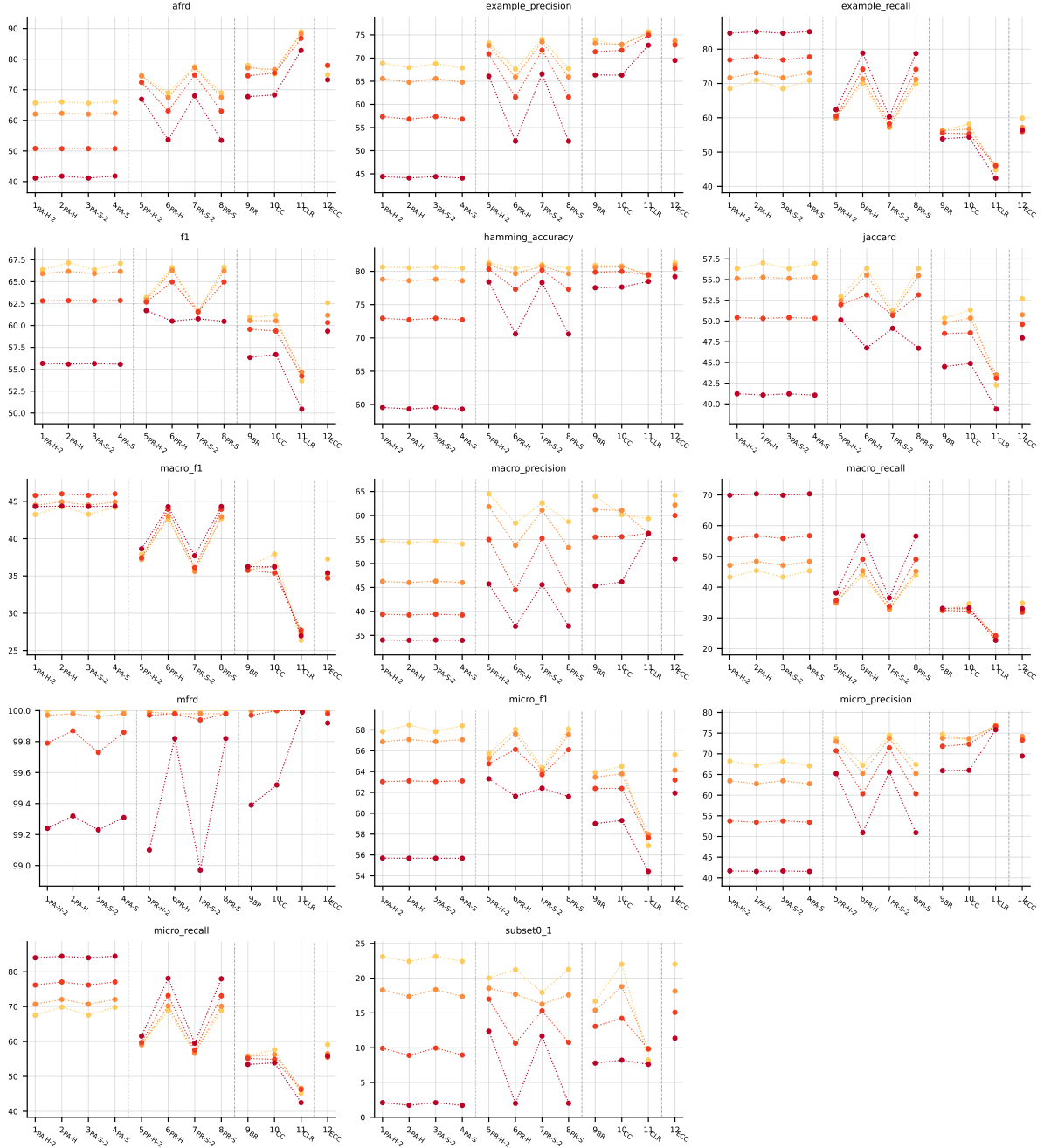


Table 20: Standard MLC predictions (BinaryVector) on Yeast (RF base learner).

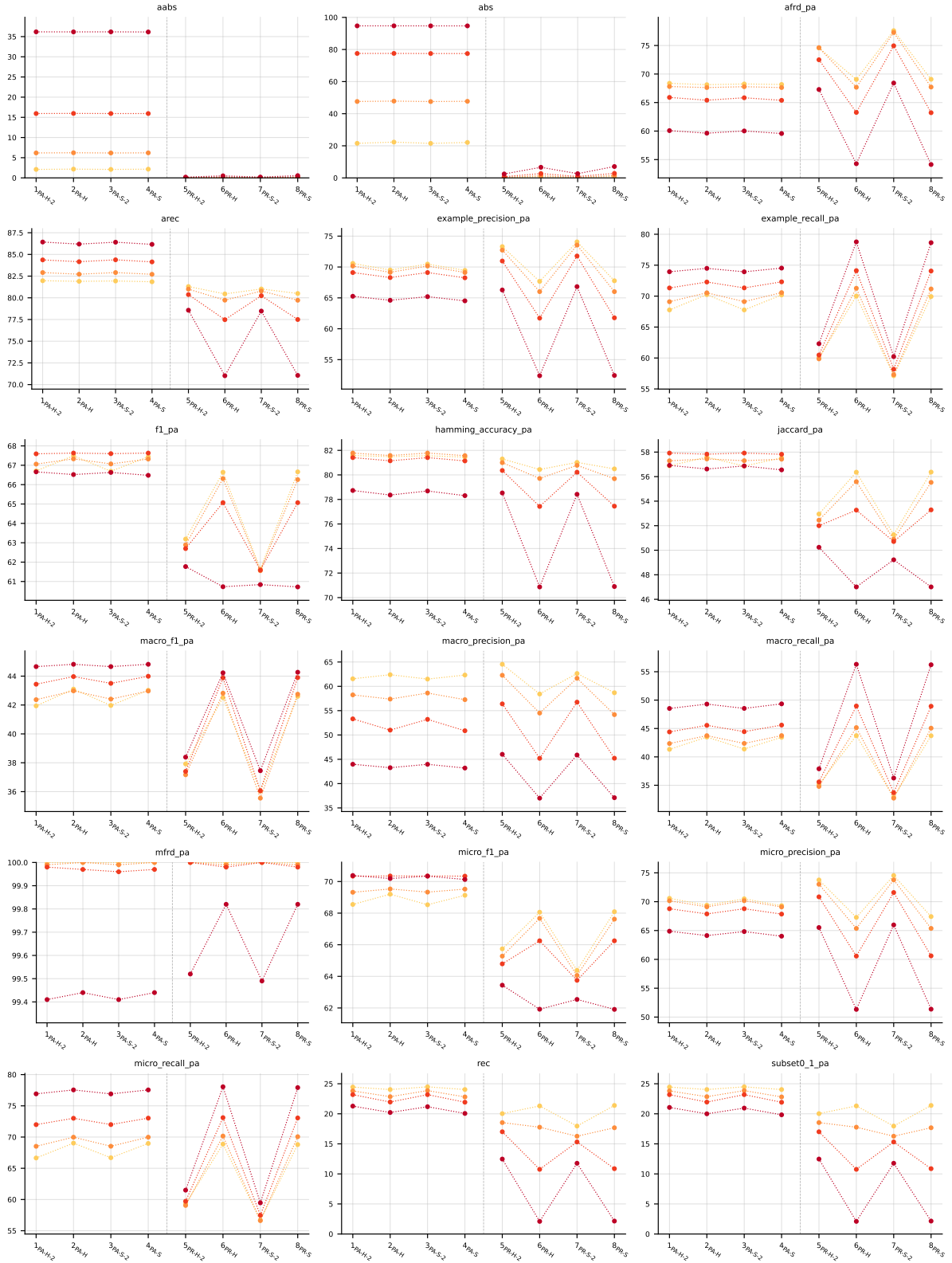


Table 21: Partial abstention (PartialAbstention) on Yeast (RF base learner).

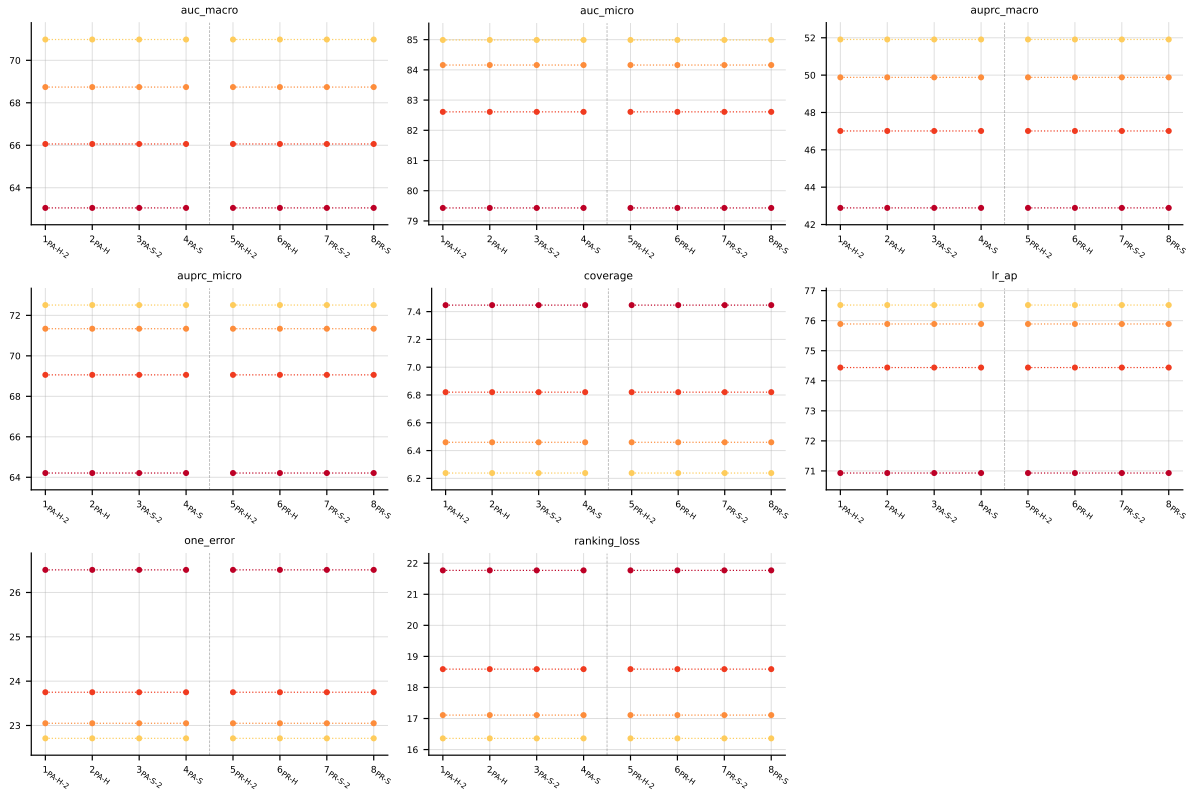


Table 22: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on Yeast (RF base learner).



Table 23: Standard MLC predictions (BinaryVector) on birds (RF base learner).

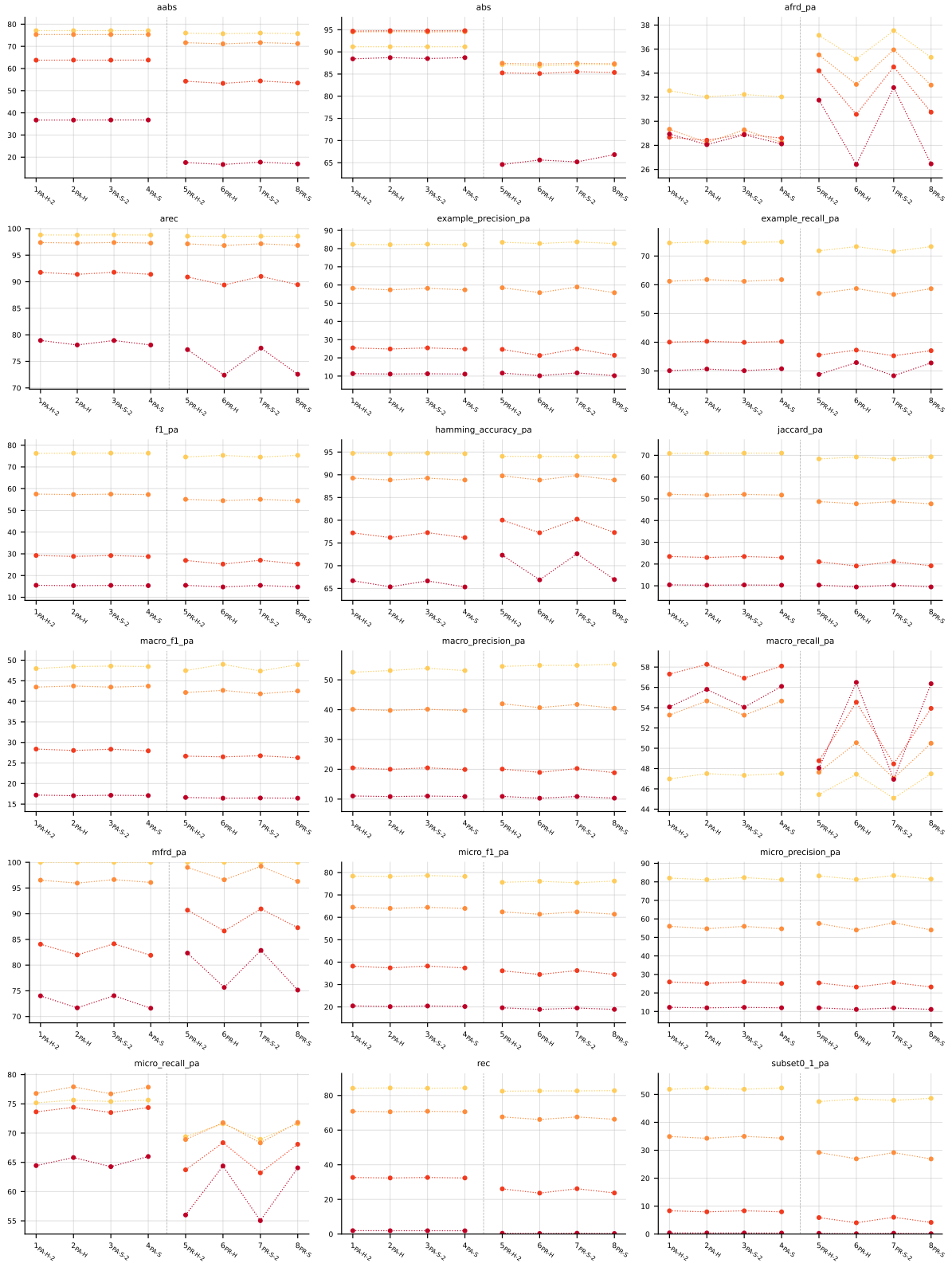


Table 24: Partial abstention (PartialAbstention) on birds (RF base learner).

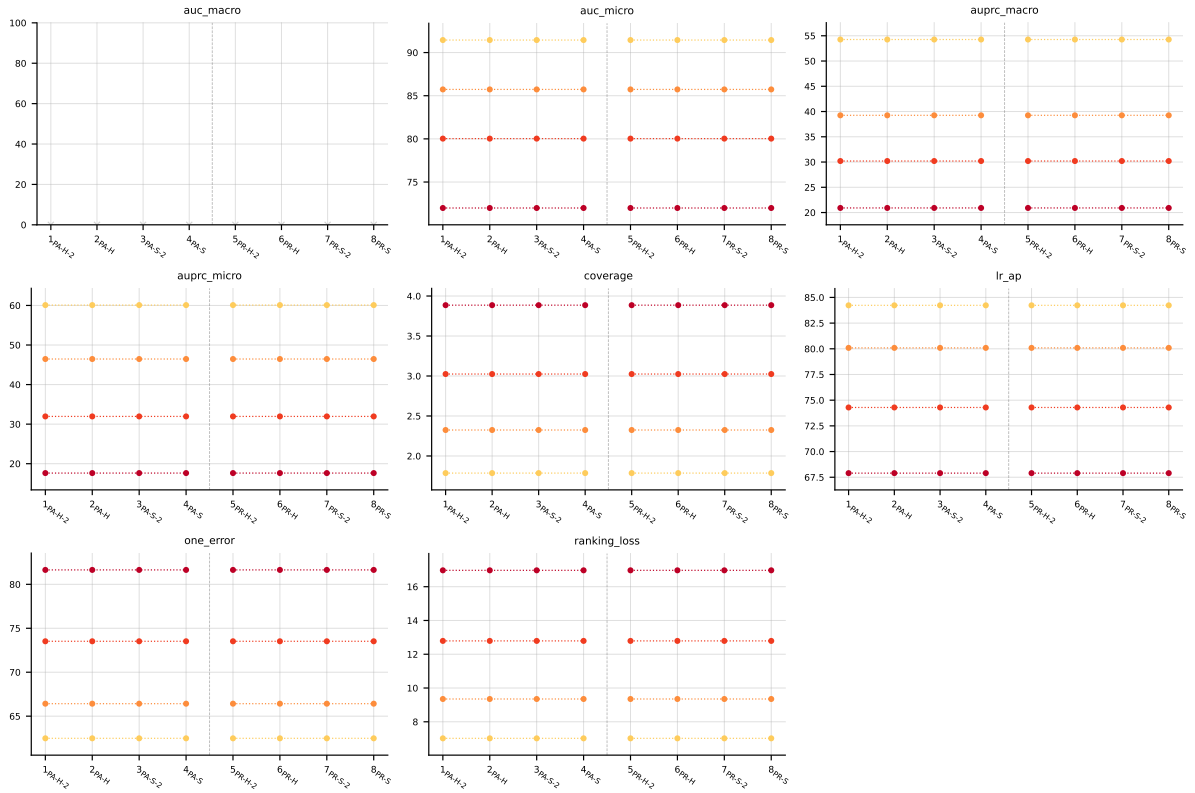


Table 25: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on **birds** (RF base learner).

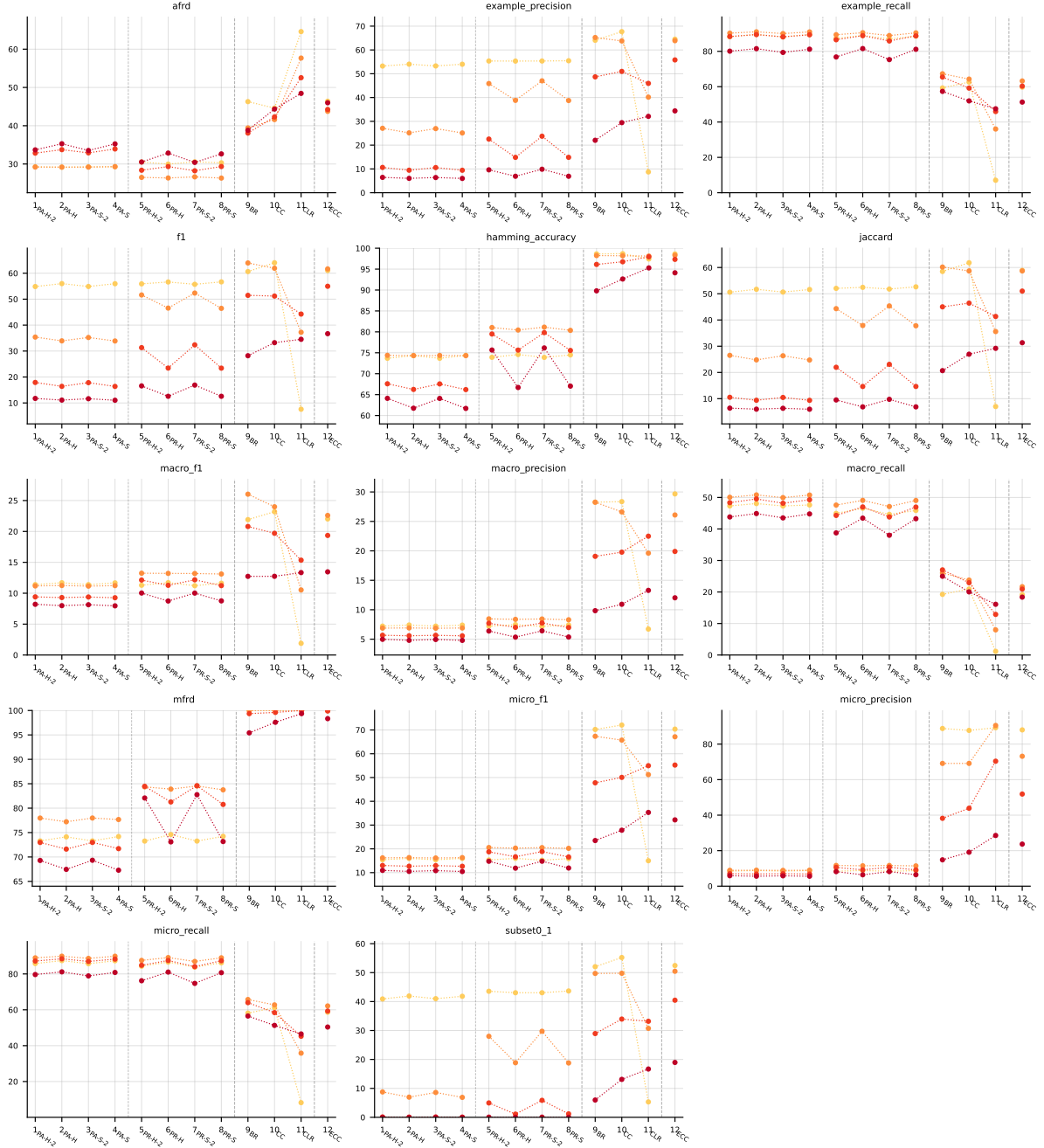


Table 26: Standard MLC predictions (BinaryVector) on medical (RF base learner).

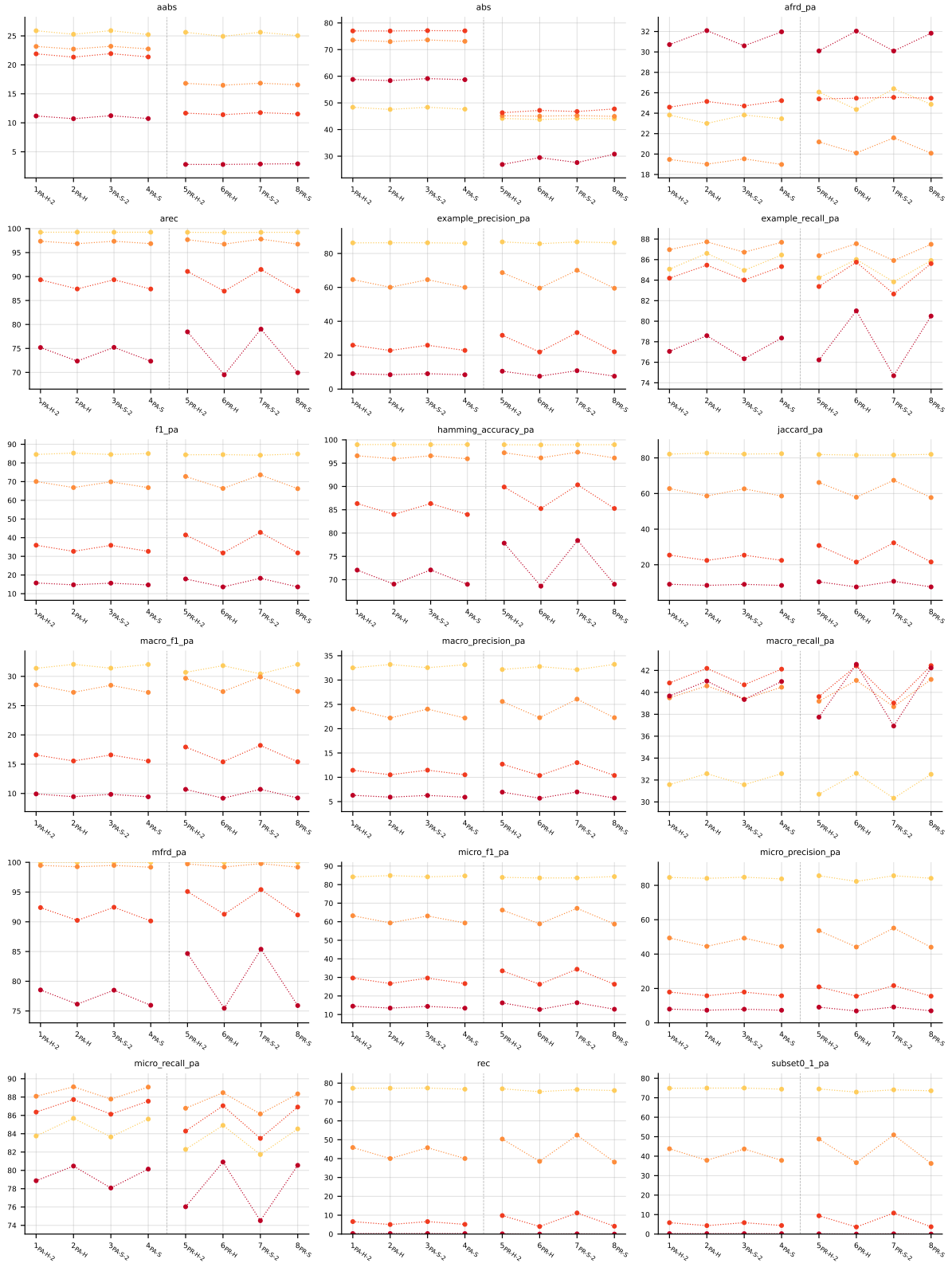


Table 27: Partial abstention (PartialAbstention) on medical (RF base learner).

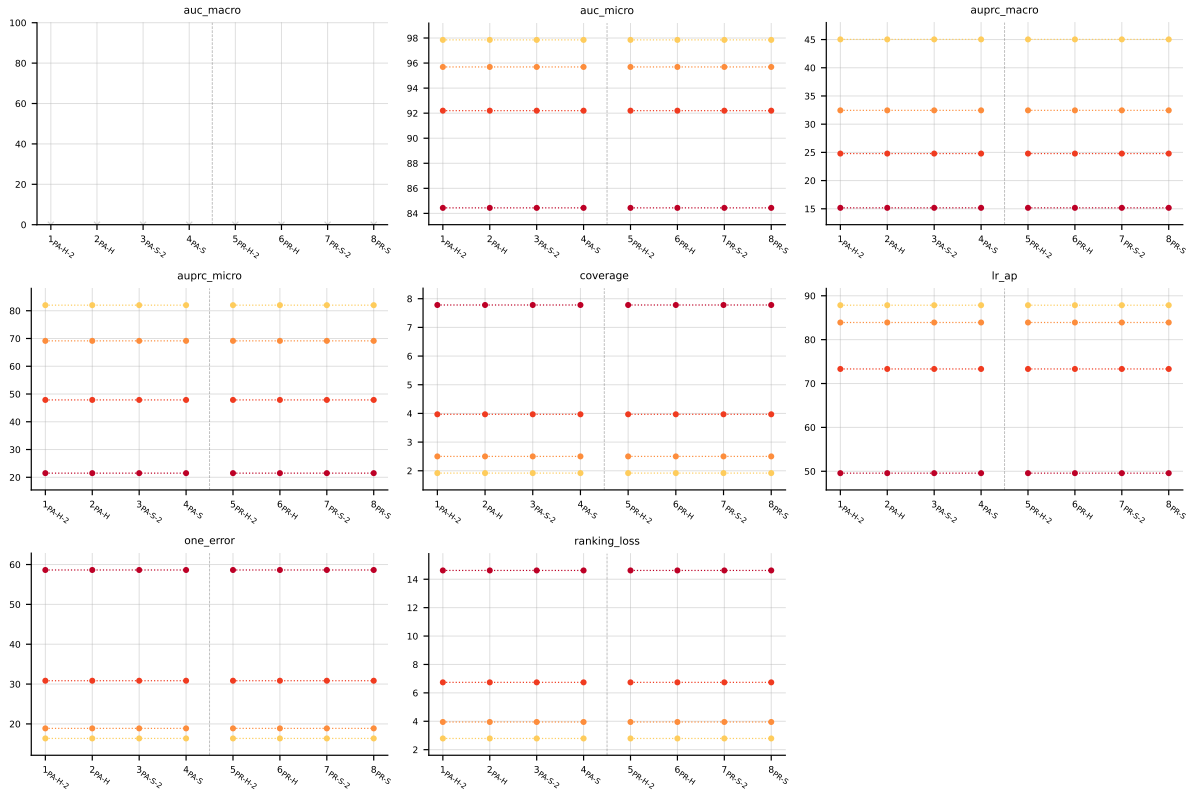


Table 28: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on `medical` (RF base learner).

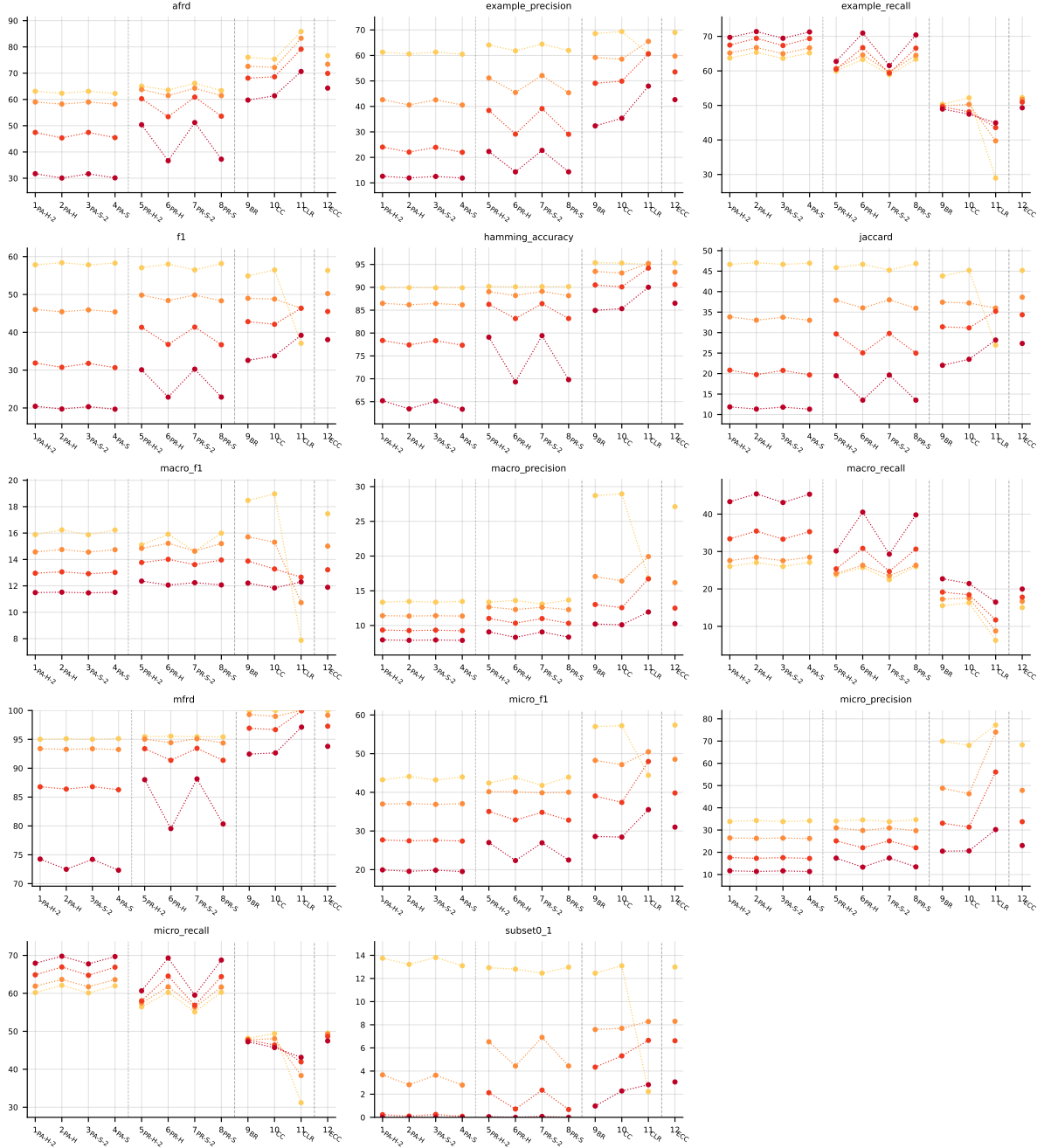


Table 29: Standard MLC predictions (BinaryVector) on **enron** (RF base learner).

A.2 Emotions ($K = 6$)

RF • emotions • BinaryVector — Standard MLC predictions

RF • emotions • PartialAbstention — Partial abstention

RF • emotions • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

A.3 Scene ($K = 6$)

RF • scene • BinaryVector — Standard MLC predictions

RF • scene • PartialAbstention — Partial abstention

RF • scene • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

A.4 PlantPseAAC ($K = 12$)

RF • PlantPseAAC • BinaryVector — Standard MLC predictions

RF • PlantPseAAC • PartialAbstention — Partial abstention

RF • PlantPseAAC • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

A.5 HumanPseAAC ($K = 14$)

RF • HumanPseAAC • BinaryVector — Standard MLC predictions

RF • HumanPseAAC • PartialAbstention — Partial abstention

RF • HumanPseAAC • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

A.6 Yeast ($K = 14$)

RF • Yeast • BinaryVector — Standard MLC predictions

RF • Yeast • PartialAbstention — Partial abstention

RF • Yeast • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

A.7 birds ($K = 19$)

RF • birds • BinaryVector — Standard MLC predictions

RF • birds • PartialAbstention — Partial abstention

RF • birds • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

A.8 medical ($K = 45$)

RF • medical • BinaryVector — Standard MLC predictions

RF • medical • PartialAbstention — Partial abstention

RF • medical • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

A.9 enron ($K = 53$)

RF • enron • BinaryVector — Standard MLC predictions

RF • enron • PartialAbstention — Partial abstention

RF • enron • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

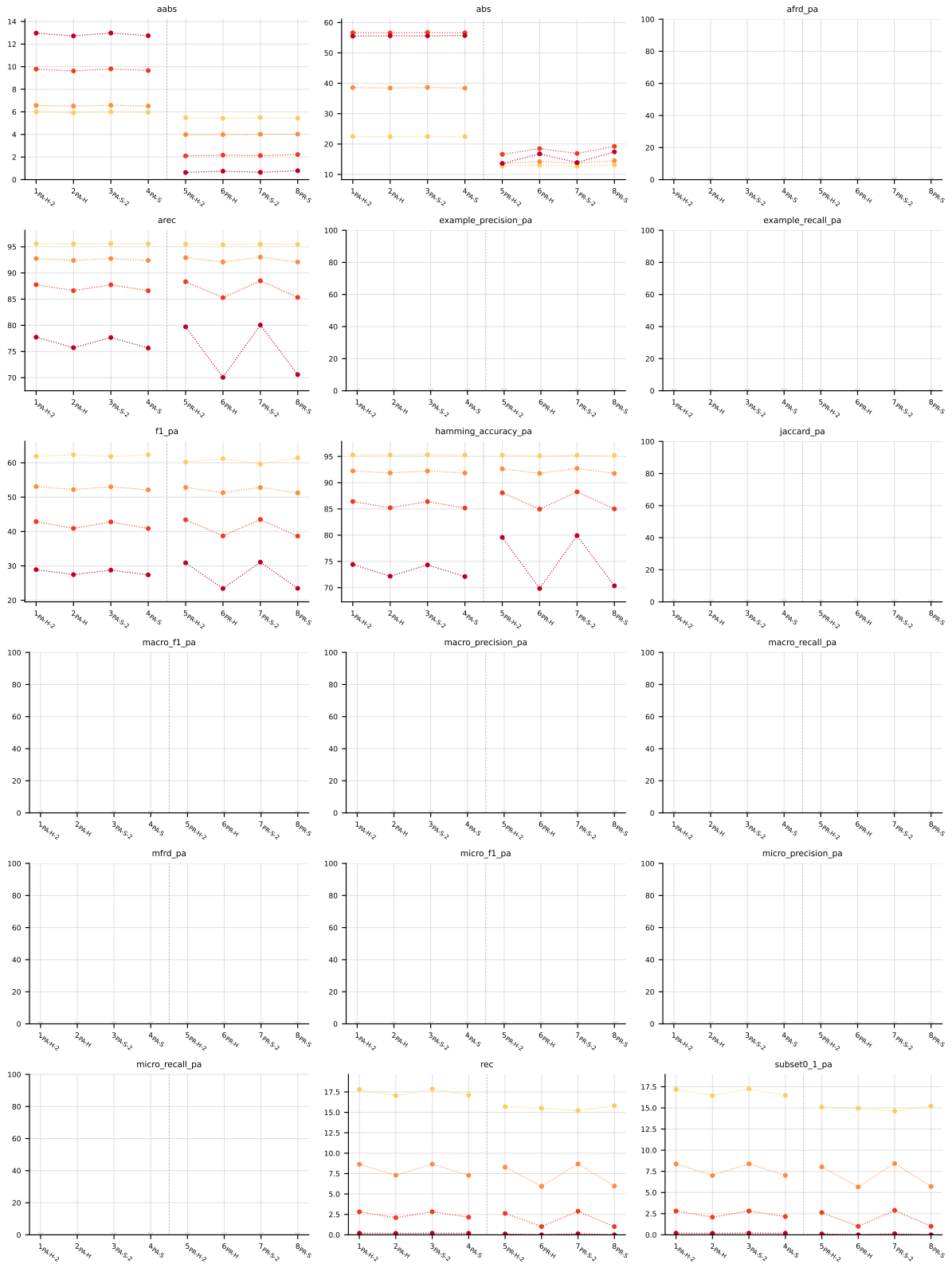


Table 30: Partial abstention (PartialAbstention) on enron (RF base learner).

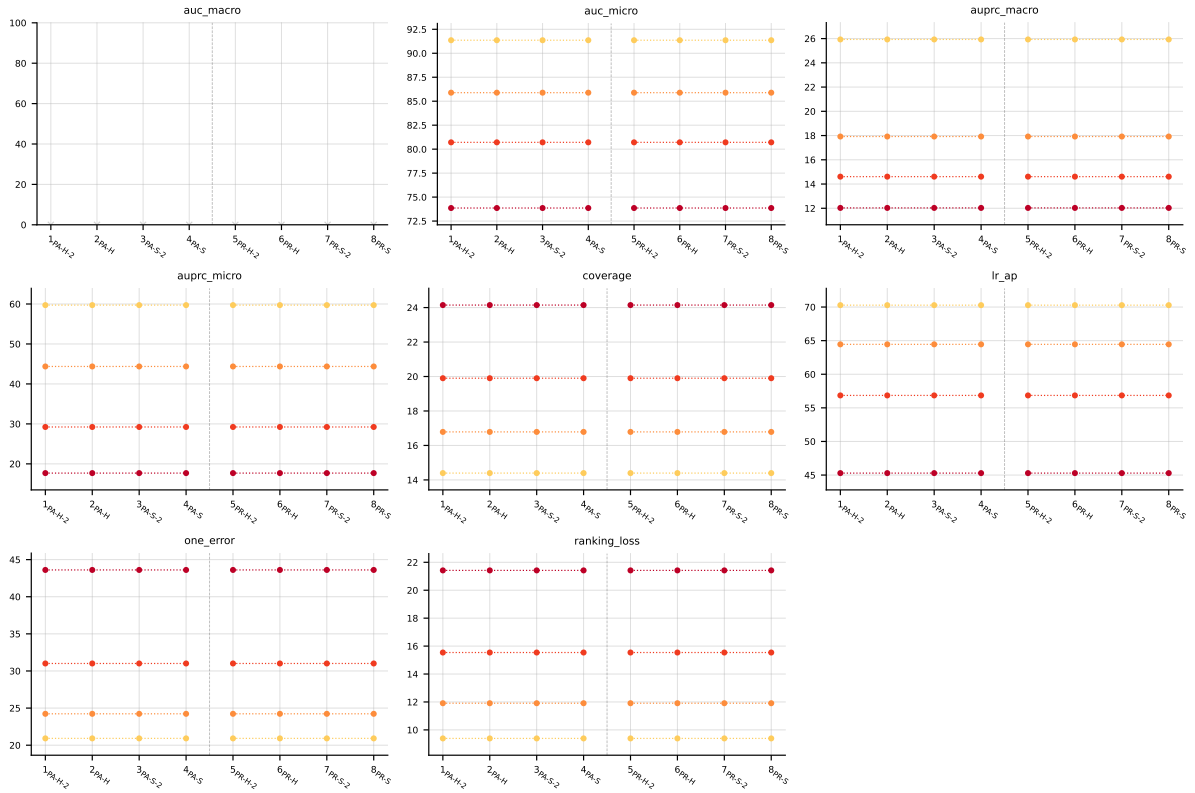


Table 31: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on **enron** (RF base learner).

B LGBM aggregated results

LGBM • Aggregate • BinaryVector — Standard MLC predictions

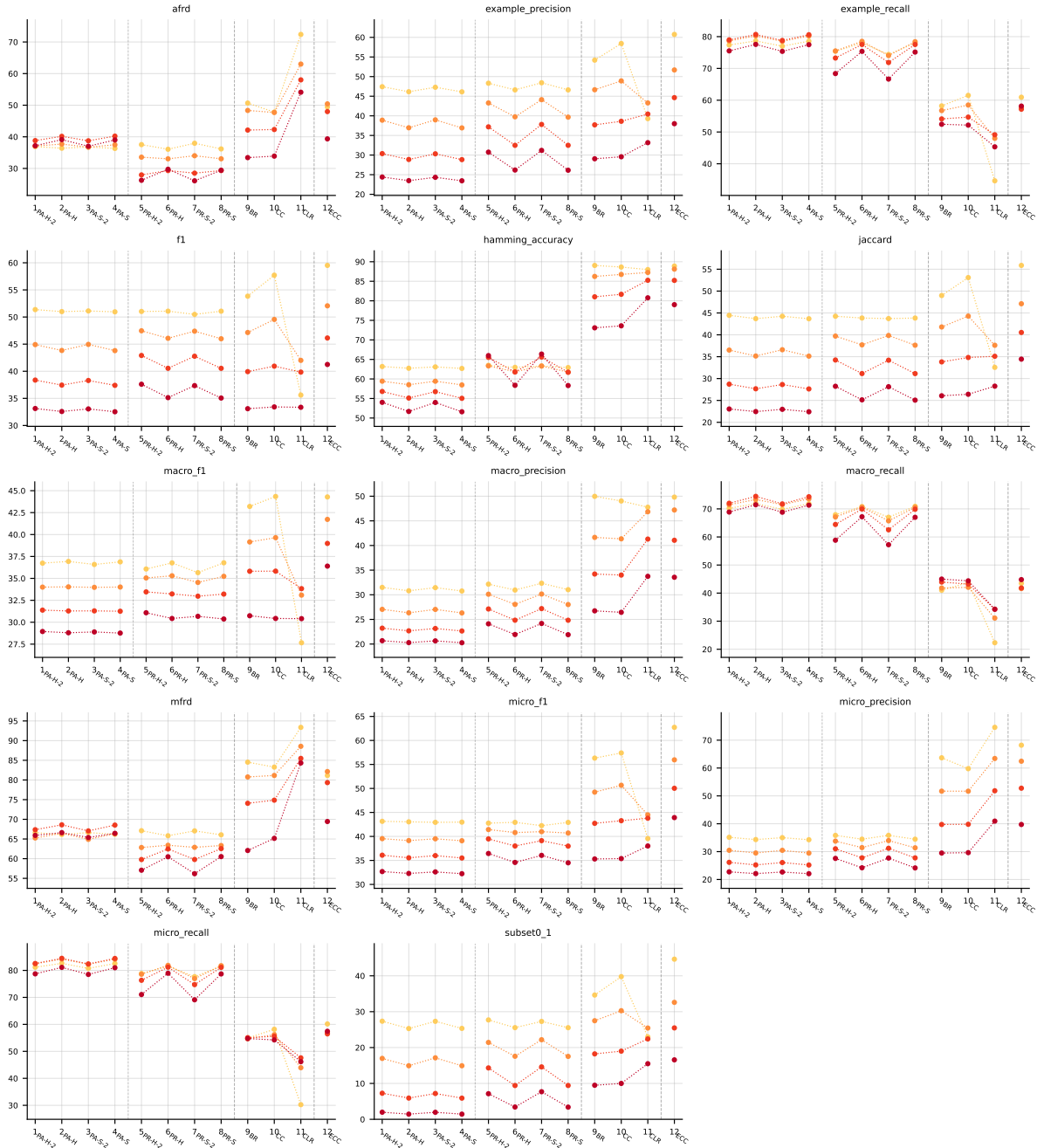


Table 32: Standard MLC predictions (BinaryVector), LGBM base learner, averaged across datasets (8 of 9: enron LGBM still training). Method indices and line colors follow the encoding paragraph above.

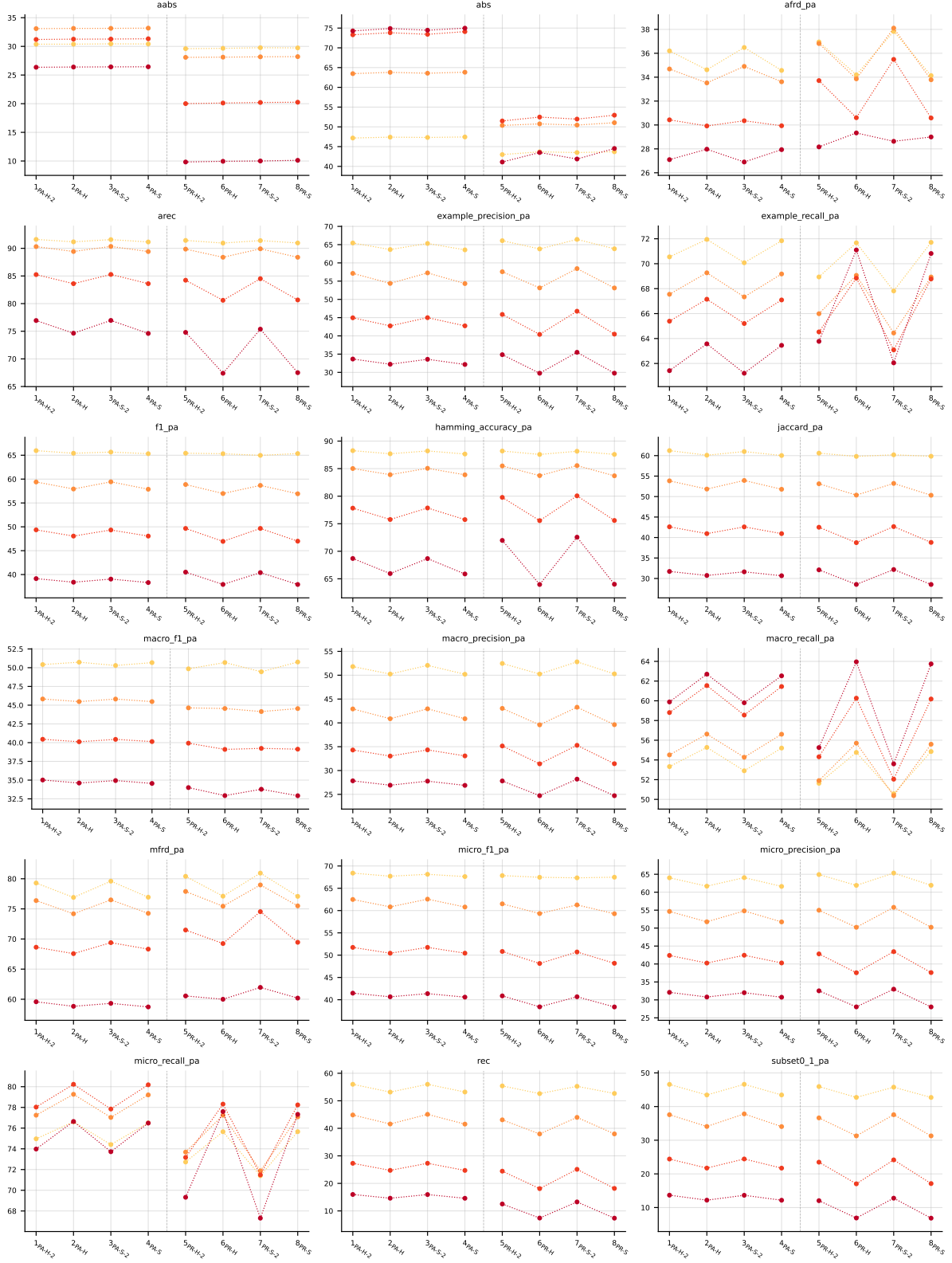


Table 33: Partial abstention (PartialAbstention), LGBM base learner, averaged across datasets (8 of 9: enron LGBM still training). Method indices and line colors follow the encoding paragraph above.

LGBM • Aggregate • PartialAbstention — Partial abstention

LGBM • Aggregate • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

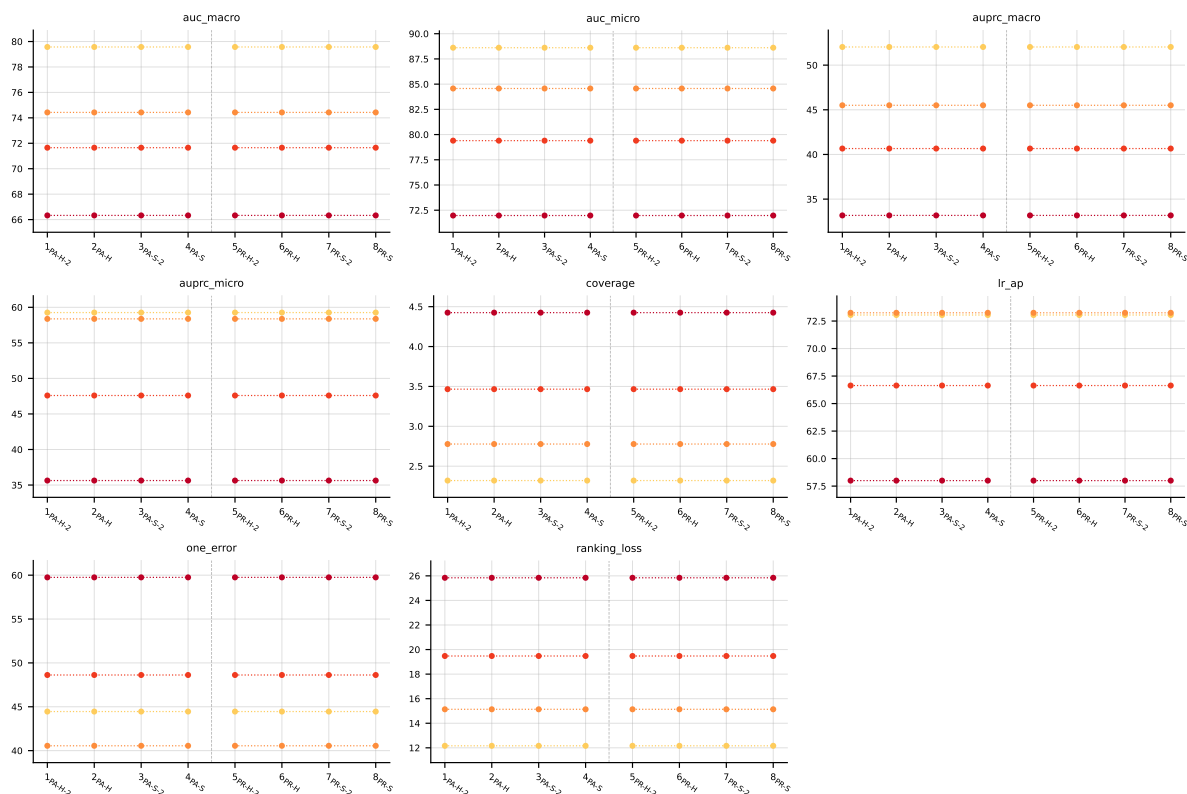


Table 34: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector), LGBM base learner, averaged across datasets (8 of 9: enron LGBM still training). Method indices and line colors follow the encoding paragraph above.